Boundaries of Stationary Feature Learning: A Minimax Barrier for Scaling Laws and the Necessity of Compositional Structure

Ivan Drozdov

Abstract

I do not derive the Chinchilla scaling law; I map the boundaries of the regime in which such a derivation could even be attempted. Working in the μP feature-learning setting on a Sobolev-on-manifold data model, I establish what the *stationary* limit of feature learning can and cannot do. **(i) A barrier.** The classical Sobolev minimax lower bound makes $\beta_0 = 2s/(2s + d^*)$ an unconditional ceiling for any estimator from D samples; feature learning is a special case, so no stationary first-order method can exceed it. **(ii) Self-organised criticality.** Treating the target's intrinsic smoothness as a *free* parameter t, the variational attractor realises source exponent $r(\nu) = t(\nu+1)/(1+2t)$ relative to its own kernel; this is monotone in the capacity exponent ν, equals exactly $r = \frac{1}{2}$ at $\nu = 1/(2t)$, and the barrier forbids the corresponding $\beta > \beta_0$ for over-aligned ν. The stationary attractor therefore self-tunes to the boundary of $\mathcal{H}^s$ and cannot stationarily overshoot it. **(iii) H1 and H2 as objects, not assumptions.** I *derive* the capacity penalty $\sum_k \lambda_k^\nu$ as the rich-regime implicit bias of a depth-L diagonal/deep-linear network, with $\nu = 1/L$, and identify the load-bearing value $\nu = d^*/(2s)$ as the depth–smoothness matching at which the attractor saturates the barrier; and I show that the previously hypothesised sublinear mode count is *not justified for Sobolev targets* (nonlinear and linear widths coincide), so the correct approximation exponent is $\alpha = 2s/d^*$ and $\alpha > \beta_0$ holds unconditionally. **(iv) Where deviations come from.** Any empirical $\beta > \beta_0$, or any acceleration of α, must originate *outside* stationary Sobolev learning: from reduced effective dimension $d^*_{\text{loc}} \ll d^*$ (compositional / Besov data structure) or from transient, non-stationary kernel alignment. This localises any empirical $\beta > \beta_0$ or acceleration of α to effects *outside* stationary Sobolev learning: reduced effective dimension $d^*_{\text{loc}} \ll d^*$ (compositional / Besov data structure) or transient, non-stationary kernel alignment. The framework complements the dynamical feature-learning models of Bordelon, Atanasov and Pehlevan: where they show how dynamics improve scaling, I prove the stationary limit cannot, which identifies exactly which mechanisms produce the improvement. I give a regime map of the $(\log N, \log D)$ plane and a decisive experimental protocol.

Index Terms

scaling laws, feature learning, mean-field theory, implicit regularisation, Sobolev spaces, minimax rates, spectral methods, compositional functions.

I. Introduction

The test loss of large neural networks improves predictably with parameters N and tokens D. Hoffmann *et al.* [1] found that for a fixed compute budget $C \propto ND$ the optimal allocation is approximately balanced and the loss surface is well fit by

$$L(N, D) \;=\; \frac{c_1}{N^\alpha} + \frac{c_2}{D^\beta} + L_0, \tag{1}$$

with $\alpha \approx 0.34$ and $\beta \approx 0.28$. Explaining (1) from first principles is an open theoretical problem.

What caught my attention was not that the law holds but that it breaks where it should not. Running kernel ridge regression sweeps on a planted Sobolev spectrum with $s/d^* \approx 0.17$, I noticed that the source exponent r stubbornly refused to exceed $\frac{1}{2}$ across widths—even when I manually forced a more aligned initialisation. The ceiling was not a numerical artefact; it matched a minimax bound I had not thought to check. That observation motivated the entire regime analysis below.

Manuscript submitted for review. Correspondence: research@drozdov.space.

A. What this paper claims, and what it does not

Earlier attempts—including an earlier version of this work—tried to *derive* (1) by positing a feature-learning mechanism with favourable exponents. That approach is circular: the favourable exponents were built into the very hypotheses meant to produce them, so the derivation recovered only what was assumed. This paper takes the opposite stance. I treat the stationary limit of feature learning as an object to be *bounded*, and I ask which features of (1) are inside those bounds and which provably are not.

Our central message is a regime statement, not a fit: *stationary feature learning on a Sobolev target is pinned to the Sobolev minimax rate β_0 and to the lazy approximation rate $\alpha = 2s/d^*$; therefore any genuinely feature-learning content of the empirical scaling law—any acceleration of α, any $\beta > \beta_0$, any task-dependence of the exponents—must arise from a departure from this regime*, namely (a) reduced effective dimension $d^*_{\text{loc}} \ll d^*$ due to compositional / hierarchical data structure, or (b) transient, non-stationary kernel alignment during training. Scaling, in this view, is the boundary between regimes, not a single power law of "success vs. compute."

B. Relation to prior theory

Spectral analyses of generalisation in the kernel / random-features regime established power-law learning curves from power-law spectra [3], [4], but the kernel there is frozen at initialisation. The feature-learning regime (finite-width networks under μP [6]) differs in that the kernel evolves and the representation deforms. The directly relevant recent line is that of Bordelon, Atanasov and Pehlevan: a solvable *dynamical* model of neural scaling laws that separates the model-size and training-time exponents and predicts an asymmetric compute-optimal rule [10], and an explicit demonstration that feature learning can *improve* scaling laws beyond the kernel limit in a task-dependent way [11]. Our results are complementary to theirs and, I believe, clarifying: where they exhibit *how* dynamics improve scaling, I prove that the *stationary* limit *cannot*, which localises the source of any improvement to exactly the dynamical and structural mechanisms their models exhibit. I also draw on the implicit-bias theory of deep linear and diagonal networks [12]–[14] to derive the capacity penalty that previous work assumed, and on classical nonlinear-approximation theory [15], [16] to correct the approximation exponent.

C. Contributions and their logical status

- **R1 (Theorem, unconditional).** The Sobolev minimax bound makes $\beta_0 = 2s/(2s + d^*)$ a ceiling for *every* estimator on a worst-case $\mathcal{H}^s$ ball; feature learning included (section VI).
- **R2 (Theorem).** With the target smoothness t free, the stationary attractor realises $r(\nu) = t(\nu + 1)/(1 + 2t)$, monotone in ν, saturating $r = \frac{1}{2}$ at $\nu = 1/(2t)$; the barrier forbids $r > \frac{1}{2}$ stationarily. Self-organised criticality (section V).
- **R3 (Theorem for diagonal/deep-linear; open in general).** The capacity penalty $\sum_k \lambda_k^\nu$ is the rich-regime implicit bias of a depth-L network with $\nu = 1/L$; "H1" ($\nu = d^*/(2s)$) is the depth–smoothness matching that saturates the barrier (section IV).
- **R4 (Theorem).** On Sobolev targets, adaptivity yields no rate gain, so $\alpha = 2s/d^*$ and $\alpha > \beta_0$ *always*; the previously hypothesised sublinear mode count is suboptimal there. The genuine gain requires $d^*_{\text{loc}} < d^*$ (section VIII).
- **Regime map and synthesis.** The $(\log N, \log D)$ plane splits at $D = N^{(2s+d^*)/d^*}$ into data-limited and architecture-limited regions (section IX); I explain precisely why stationary learning cannot account for the empirical exponents and give the test that decides the operative mechanism (section X).

I use the labels *H1* and *H2* throughout, but their role has changed: they are not assumptions the reader must grant, they are objects whose validity boundaries I map. H1 has its form derived and its critical value characterised (R3); H2 is shown to fail as originally stated and is replaced by the correct effective-dimension picture (R4).

II. Setup and Notation

A. Data manifold and function class

Input data $x \in \mathbb{R}^d$ lie on a compact Riemannian manifold $\mathcal{M} \subset \mathbb{R}^d$ of intrinsic dimension $d^* \ll d$, with Laplace–Beltrami eigendecomposition $-\Delta_{\mathcal{M}}\psi_k = \mu_k\psi_k$, $0 = \mu_0 < \mu_1 \le \cdots$, and $\{\psi_k\}$ an $L^2(\mathcal{M})$-orthonormal basis. By Weyl's law $\mu_k \asymp k^{2/d^*}$. The order-s Sobolev kernel has eigenvalues $\sigma_k \asymp k^{-b_0}$ with

$$b_0 := \frac{2s}{d^*} \qquad \text{(Sobolev capacity exponent)}. \tag{2}$$

The target $f^* = \sum_k \bar{a}_k\psi_k$, $\bar{a}_k = \langle f^*, \psi_k\rangle_{L^2}$, lies in $\mathcal{H}^s(\mathcal{M})$, i.e. $\sum_k(1+\mu_k)^s\bar{a}_k^2 < \infty$. Crucially, I *do not* fix the coefficient profile to the border of $\mathcal{H}^s$. I treat the target's intrinsic spectral decay as a *free parameter*:

$$\bar{a}_k^2 \asymp k^{-(1+2t)}, \qquad t > 0, \tag{3}$$

where t is the source exponent of f^* relative to the Sobolev kernel. The border-$\mathcal{H}^s$ profile is the special case $t = s/d^* = b_0/2$. Treating t as free, rather than locking it to the architecture, is the single change from the earlier draft; see theorem V.3 for the technical reason.

B. Two source exponents, kept distinct

The estimation rate is governed by the source exponent of f^* relative to the kernel the estimator *uses*. Two kernels appear:

- the fixed Sobolev/geometry kernel $\sigma_k \asymp k^{-b_0}$ (the lazy / NTK basis), against which f^* has source exponent t/b_0;
- the learned stationary attractor $\lambda_k^* \asymp k^{-b}$ produced by feature learning, against which f^* has source exponent r.

Conflating these two is what produced the earlier tautology. I keep them separate throughout.

C. Loss and architecture

The loss is squared error $\mathcal{L}(f) = \mathbb{E}_{(x,y)}[(f(x)-y)^2]$; cross-entropy alters the gradient structure and is outside our scope. I consider an L-layer width-n network under the maximal-update parametrisation (μP),

$$\begin{aligned} \mathbf{h}^{(0)} = x, \qquad \mathbf{h}^{(\ell)} &= \tfrac{1}{\sqrt{n}}\mathbf{W}^{(\ell)}\phi(\mathbf{h}^{(\ell-1)}), \quad \ell = 1,\dots,L, \\ f(x;\theta) &= \tfrac{1}{\sqrt{n}}\mathbf{v}^\top\phi(\mathbf{h}^{(L)}), \end{aligned} \tag{4}$$

ϕ smooth bounded with $\phi'' \in L^\infty$, $N = \Theta(n^2L)$, and learning rate $\eta = \Theta(1)$ so that $\Delta\mathbf{W}^{(\ell)} = \Theta(1)$—the feature-learning hallmark.

III. Act I: Mean-Field Closure of the Evolving Kernel

Define the layer-ℓ covariance $K^{(\ell)}(x,x';t) = \mathbb{E}[\phi(\mathbf{h}^{(\ell)}(x;t))\,\phi(\mathbf{h}^{(\ell)}(x';t))^\top]$.

Proposition III.1 (Mean-field closure; Yang–Hu)**.** *In the limit $n \to \infty$ at fixed depth L, the Master Theorem of Tensor Programs IV [6] guarantees that the pre-activation fields converge weakly to a deterministic Gaussian process, under which the dynamics of $K^{(\ell)}$ close into*

$$\frac{\partial K^{(\ell)}}{\partial t} = \dot{\phi}^{\otimes 2} \star K^{(\ell)} - \eta\left(K^{(\ell)}\hat{\Sigma}_D^{-1}\nabla_K\mathcal{L}\right), \tag{5}$$

with $\hat{\Sigma}_D = D^{-1}\sum_i\phi(x_i)\phi(x_i)^\top$ and Stein–Price convolution $\dot{\phi}^{\otimes 2} \star K$ legitimate by Gaussianity. At $\nabla_K\mathcal{L} = 0$ this reduces to the NTK equation.

I state theorem III.1 as a closure, not as a new theorem: it is an application of [6], and I flag that the order of the $n \to \infty$ and $L \to \infty$ limits is not interchangeable, so any large-L statement is heuristic. Projecting onto $\{\psi_k\}$, $K^{(L)} = \sum_k \lambda_k(t)\psi_k \otimes \psi_k$, gives the spectral system

$$\frac{d\lambda_k}{dt} = -\eta \frac{\partial V}{\partial \lambda_k} + \Gamma_k[\{\lambda_j\}_{j\neq k}, t], \tag{6}$$

with Γ_k the inter-mode coupling, $|\Gamma_k| = O(k^{-\gamma})$, and V the effective potential. The *form* of V—in particular its capacity term—is no longer postulated here; it is derived in section IV.

IV. The Capacity Penalty as Implicit Bias (Object H1)

For the diagonal/deep-linear class, H1 is now a derived object, not a modelling choice.

Lemma IV.1 (Reconstruction term)**.** *In the learned eigenbasis the population squared loss contributes the modewise reconstruction cost $\sum_k \bar{a}_k^2/\lambda_k$: representing coefficient $\bar{a}_k$ with feature scale λ_k costs RKHS energy $\bar{a}_k^2/\lambda_k$. This term is forced, not chosen.*

Lemma IV.2 (Capacity term from depth; rich-regime implicit bias)**.** *Consider the standard tractable proxy for μP feature learning: a depth-L diagonal linear (equivalently deep matrix-factorisation) network in which mode k is parametrised by a product $w_k = \prod_{j=1}^{L} \theta_{k,j}$ and the learned kernel eigenvalue is the feature second moment $\lambda_k \asymp w_k^2$. Gradient flow from vanishing balanced initialisation converges, among interpolators, to the minimiser of the Schatten quasinorm $\sum_k |w_k|^{2/L}$ [12]–[14]. In terms of $\lambda_k = w_k^2$ this coincides with a capacity penalty*

$$\sum_k \lambda_k^{\nu}, \qquad \boxed{\nu = \tfrac{1}{L}}. \tag{7}$$

Together theorems IV.1 and IV.2 *derive* the potential

$$V(\boldsymbol{\lambda}) = \sum_{k\geq 1} \frac{\bar{a}_k^2}{\lambda_k} + \mu \sum_{k\geq 1} \lambda_k^{\nu}, \qquad \mu, \nu > 0, \tag{8}$$

as the Lagrangian of $\min_{\boldsymbol{\lambda}} \sum_k \bar{a}_k^2/\lambda_k$ subject to the implicit Schatten budget $\sum_k \lambda_k^{1/L} \leq C$.

Theorem IV.3 (H1 as the minimax-saturating depth)**.** *For the diagonal/deep-linear class, $\nu = 1/L$. The load-bearing value $\nu = d^*/(2s)$ used as "Hypothesis H1" is precisely the matching condition*

$$\frac{1}{L} = \frac{1}{2t} = \frac{d^*}{2s} \iff L = \frac{2s}{d^*} = b_0, \tag{9}$$

i.e. the depth at which the network's implicit Schatten exponent equals the target's intrinsic smoothness scale. By theorem V.2 below this is the unique depth at which the stationary attractor saturates the Sobolev minimax barrier.

Remark IV.4 (Scope and the one remaining gap)**.** *Equation* (7) *is a theorem for the diagonal/deep-linear rich regime. Extending $\nu = 1/L$ to general nonlinear μP networks—proving that the tensor-program dynamics induce the same depth-set Schatten penalty—is the single open step (section XI-B). What section IV buys is decisive in kind, not degree: H1's* form *is no longer a modelling choice, and its value has a mechanism. A realisability caveat: $L = 2s/d^*$ is an integer depth only for smooth targets ($s \geq d^*$); the rough regime $s \ll d^*$ cannot be matched by a finite-depth stationary network, foreshadowing section X.*

V. Act II: The Stationary Attractor and Self-Organised Criticality

The KKT stationarity condition $\partial V/\partial\lambda_k = 0$ on (8) gives $\bar{a}_k^2 = \mu\nu(\lambda_k^*)^{\nu+1}$, so

$$\lambda_k^* \propto (\bar{a}_k^2)^{1/(\nu+1)} \asymp k^{-b}, \qquad b := \frac{1+2t}{\nu+1}. \tag{10}$$

Local asymptotic stability follows, under the assumption that Γ_k is a gradient $\Gamma_k = -\eta\,\partial\Phi/\partial\lambda_k$, from LaSalle's principle applied to $V_{\text{tot}} = V + \Phi$, whose Hessian is diagonally dominated near $\boldsymbol{\lambda}^*$ (the reconstruction curvature $2\bar{a}_k^2/(\lambda_k^*)^3$ dominates the summable off-diagonal coupling). I claim local, not global, stability.

Theorem V.1 (Realised source exponent and its ceiling)**.** *With the free target* (3) *and attractor* (10)*, the source exponent of f^* relative to the attractor operator is*

$$r(\nu) = \sup\Big\{r \geq 0 : \textstyle\sum_k \bar{a}_k^2/(\lambda_k^*)^{2r} < \infty\Big\} = \frac{t}{b} = \frac{t(\nu+1)}{1+2t}. \tag{11}$$

The map $\nu \mapsto r(\nu)$ is strictly increasing and $r(\nu) \leq \frac{1}{2} \iff \nu \leq 1/(2t)$, with $r(1/(2t)) = \frac{1}{2}$.

Proof. $\sum_k \bar{a}_k^2/(\lambda_k^*)^{2r} \asymp \sum_k k^{-(1+2t)+2rb}$ converges iff $(1+2t) - 2rb > 1$, i.e. $r < t/b$; substitute $b = (1+2t)/(\nu+1)$. Solving $r(\nu) = \frac{1}{2}$ yields $2t\nu = 1$, i.e. $\nu = 1/(2t)$. □

Theorem V.2 (Self-organised criticality)**.** *Combining theorem V.1 with the barrier theorem VI.1, for a worst-case $\mathcal{H}^s$ target ($t = s/d^*$):*

(i) at the matched capacity $\nu_\star = 1/(2t) = d^*/(2s)$, the attractor has $r = \frac{1}{2}$ and a spectral estimator on it attains $\mathcal{E}_{\text{est}} \asymp D^{-\beta_0}$: stationary feature learning is minimax-rate optimal;

(ii) for $\nu > \nu_\star$, (11) *would predict $r > \frac{1}{2}$ and $\beta(r) > \beta_0$, but theorem VI.1 forbids this on the worst-case ball, so an over-aligned stationary kernel is misspecified and its realised rate reverts to β_0.*

Thus $r = \frac{1}{2}$ is the unique saturation point: stationary feature learning self-tunes to the boundary of $\mathcal{H}^s$ and cannot stationarily overshoot it.

Remark V.3 (No circularity: two independent parameters)**.** *The earlier draft fixed both the border profile* and *ν, so $r = \frac{1}{2}$ was an identity. Here t is a free property of the data and $\nu = 1/L$ is a property of the architecture; $r = \frac{1}{2}$ is the* coincidence *of the two, and theorem V.1 exhibits the whole family $r(\nu)$ around it. The content is a ceiling on a tunable knob, not a tautology.*

VI. The Unconditional Barrier (R1)

Theorem VI.1 (Sobolev minimax barrier)**.** *Let f^* range over $\{\|f\|_{\mathcal{H}^s(\mathcal{M})} \leq R\}$, $\dim\mathcal{M} = d^*$. For any estimator $\hat{f}_D$ from D i.i.d. samples with sub-Gaussian label noise,*

$$\inf_{\hat{f}_D}\ \sup_{\|f^*\|_{\mathcal{H}^s} \leq R} \mathbb{E}\left\|\hat{f}_D - f^*\right\|_{L^2}^2 \gtrsim D^{-\beta_0}, \qquad \beta_0 = \frac{2s}{2s+d^*}. \tag{12}$$

Hence every learning rule—lazy, feature-learning, stationary or transient—obeys $\beta \leq \beta_0$ in the worst case over the ball.

Proof sketch. Standard [4], [5]. Build a hypercube of $M \asymp \delta^{-d^*/s}$ perturbations supported near the resolution scale $k \asymp \delta^{-d^*}$, each of $\mathcal{H}^s$-norm $\leq R$ and pairwise L^2-separated by $\asymp \varepsilon$; the per-sample KL divergence is $\asymp \varepsilon^2$, and Fano forces error $\gtrsim \varepsilon^2$ when $D\varepsilon^2 \lesssim \log M$. Optimising gives $\varepsilon^2 \asymp D^{-2s/(2s+d^*)}$. □

Remark VI.2. *theorem VI.1 is the load the earlier Proposition tried, circularly, to carry. It needs no attractor and no H1. Everything above and below* characterises *how feature learning sits against this barrier.*

VII. Estimation Rate and the Optimiser Ceiling

On the attractor kernel, learning the coordinates from D samples is kernel ridge regression. With $\mathcal{N}(\rho) = \sum_k \lambda_k^*/(\lambda_k^* + \rho) \asymp \rho^{-d^*/(2s)}$ (effective dimension; the integral converges for $b_0 > 1$) and source exponent r, the source/capacity rate of [4] is

$$\mathcal{E}_{\text{est}}(D) \asymp D^{-\beta(r)}, \qquad \beta(r) = \frac{2r}{2r + d^*/(2s)} = \frac{4rs}{4rs + d^*}. \tag{13}$$

At the critical $r = \frac{1}{2}$ this is $\beta_0 = 2s/(2s + d^*)$; the whole estimation question collapses to the single scalar r, which theorem V.2 pins at $\frac{1}{2}$ for stationary learning.

Proposition VII.1 (Early-stopping/ridge duality; no optimiser escape)**.** *On a fixed power-law kernel, ridge applies the spectral filter $\lambda/(\lambda + \rho)$ and gradient-flow early stopping applies $1 - e^{-\lambda\tau}$; both lie in the admissible spectral-regulariser family of qualification ≥ 1, and for $r \leq 1$ their optimal rates coincide [17], [18]. Consequently r is a property of (target, kernel), not of the optimiser: among stationary-kernel first-order methods, training time and ridge level move only along* (13)*, so $r \leq \frac{1}{2}$ and $\beta \leq \beta_0$. SGD adds an $O(D^{-1})$ variance floor but does not raise r; Adam's preconditioner rescales the metric and is therefore* not *a stationary-kernel method—any gain it produces is, by definition, kernel modification, i.e. non-stationary.*

VIII. Approximation Error: Correcting H2 (R4)

The earlier draft posited a sublinear effective-mode count $K_{\text{eff}} \asymp N^{2s/(2s+d^*)}$ to soften α, conceding that the naive entropy balance returns $K_{\text{eff}} \asymp N$. Approximation theory settles the matter.

Theorem VIII.1 (No adaptivity gain over a Sobolev ball)**.** *For the Sobolev ball $\mathcal{H}^s(\mathcal{M})$, $\dim \mathcal{M} = d^*$, the nonlinear (best N-term / continuous / manifold) width matches the linear width in rate,*

$$d_N(\mathcal{H}^s) \asymp \delta_N(\mathcal{H}^s) \asymp N^{-s/d^*} \;\Rightarrow\; \mathcal{E}_{\text{appr}}(N) \asymp N^{-2s/d^*} \;\Rightarrow\; \boxed{\alpha = \tfrac{2s}{d^*}} \tag{14}$$

[15], [16]. Adaptively choosing which features to use does not improve the worst-case rate over a Sobolev ball: the Laplacian eigenbasis is already rate-optimal for $\mathcal{H}^s$.

Corollary VIII.2 (The sublinear count is suboptimal on Sobolev)**.** *A network resolving a sublinear $K_{\text{eff}} \ll N$ achieves $\alpha = (2s/d^*)\cdot(K_{\text{eff}}$ exponent$) < 2s/d^*$—strictly worse than the lazy rate, with no compensating benefit on a Sobolev target. The rational stationary outcome on Sobolev data is the lazy count $K_{\text{eff}} \asymp N$ and $\alpha = 2s/d^*$. "H2" as previously stated is not merely unproven; it is suboptimal for the class it was applied to.*

Theorem VIII.3 (Corrected asymmetry)**.** *With $\alpha = 2s/d^*$ and $\beta_0 = 2s/(2s + d^*)$,*

$$\alpha - \beta_0 = \frac{4s^2}{d^*(2s + d^*)} > 0 \quad \textit{for all } s, d^* > 0. \tag{15}$$

Parameters are the more efficient resource unconditionally*, consistent with $\alpha > \beta$ empirically. (The expression $4s^2/[d^*(2s + d^*)]$, previously mislabelled α, is in fact the gap $\alpha - \beta_0$.)*

a) Where the genuine gain lives: $d^*_{\text{loc}} < d^*$.: Since adaptivity is rate-neutral on Sobolev, improvement requires the target to have less than ambient effective dimension. Two unified mechanisms:

- **Compositional / hierarchical targets** (the staircase regime [20]): if f^* composes pieces of local dimension $d^*_{\text{loc}} \ll d^*$, the achievable rate is $N^{-2s/d^*_{\text{loc}}}$, and feature learning's role is to *discover* d^*_{loc} [19]. The "sublinear mode count" was a confused proxy for the dimension reduction $d^* \to d^*_{\text{loc}}$.
- **Inhomogeneous / Besov targets** ($B^s_{p,q}$, $p < 2$): adaptive N-term approximation strictly beats linear, and the correct sublinear budget follows from the nonlinear-approximation rate of that class [16].

In both cases the gain is *not* stationary-Sobolev; it is the data-structure mechanism, and it is the regime where the feature-learning improvements of [11] appear.

IX. Scaling as a Regime Change

With the corrected $\alpha = 2s/d^*$, the loss decomposes as $\mathbb{E}L \asymp N^{-\alpha} + D^{-\beta_0} + L_0$, the two terms dominating disjoint regions separated by

$$N^{-\alpha} = D^{-\beta_0} \iff D_{\text{cross}} = N^{\alpha/\beta_0} = N^{(2s+d^*)/d^*}. \tag{16}$$

(The earlier $D = N^{\alpha}$ was an artifact of the sublinear K_{eff}.) The same knee arises from resolution ranks: parameters resolve $\asymp N$ modes, D samples resolve $\asymp D^{d^*/(2s+d^*)}$, and these coincide on (16). Hence "more compute" is not a uniform gain: across the knee one resource saturates while the other binds. The compute frontier $C = ND$ gives

$$N^* \propto C^{d^*/[2(s+d^*)]}, \qquad D^* \propto C^{(2s+d^*)/[2(s+d^*)]}, \tag{17}$$

allocating marginal compute data-heavily since $\alpha > \beta_0$—the qualitative asymmetry also reported by [10].

- $D \gg N^{(2s+d^*)/d^*}$: architecture-limited; $\mathcal{E}_{\text{appr}} \asymp N^{-\alpha}$ dominates; data has diminishing returns.
- $D \ll N^{(2s+d^*)/d^*}$: data-limited; spectral undersampling dominates; parameters have diminishing returns.
- $D \approx N^{(2s+d^*)/d^*}$: balanced; both errors comparable.

X. Synthesis: Why Stationary Learning Cannot Explain Chinchilla

Assembling R1–R4:

1. **It cannot exceed** β_0 (theorem VI.1, theorem VII.1). If empirical $\beta = \beta_0$, feature learning adds nothing beyond fixed-kernel KRR; if $\beta > \beta_0$, stationary learning cannot produce it.
2. **It cannot accelerate** α **on Sobolev** (theorem VIII.1): the rate is the lazy $\alpha = 2s/d^*$ unless $d^*_{\text{loc}} < d^*$.
3. **The empirical exponents miss the trace-class window.** Matching $\alpha = 2s/d^* \approx 0.34$ forces $s/d^* \approx 0.17$, hence $b_0 \approx 0.34 < 1$: *not* trace class. But the effective-dimension integral converges only for $b_0 > 1$, so the analysis yielding β_0 is invalid in the regime needed to fit α. (At $s/d^* \approx 0.17$ one still gets $\beta_0 \approx 0.25$ near the observed 0.28—close enough to place the model in the right ballpark, far enough to show it does not cleanly apply.)
4. **Separability needs** N**-independent** r**.** If feature learning made $r = r(N)$, the exponents would drift and the additive form would break; the empirical stability of α, β is mild evidence against a width-dependent stationary alignment.

Conclusion. Any feature-learning content of the empirical scaling law is explained by transient / non-stationary kernel dynamics or by reduced-effective-dimension (compositional) data structure—never by the stationary attractor, which is pinned to the Sobolev minimax barrier. This is a falsifiable claim that complements, rather than competes with, the dynamical models of [10], [11]: they show how the boundary is crossed; I show that it must be crossed.

Remark X.1 (Decisive test)**.** *Form the post-training empirical kernel, extract* $\{\hat{\lambda}_k, \hat{\psi}_k\}$*, read* b *from* $\hat{\lambda}_k \asymp k^{-b}$ *and* r *from* $\hat{a}_k^2 \asymp k^{-(2rb+1)}$*, at several widths* N *and at* D *past the knee* (16). *If* r *stays at* $\frac{1}{2}$ *across* N*: stationarity holds,* $\beta = \beta_0$*, no rate enhancement. If* r *rises: the operative kernel is transient, and the measured* r *pins* β *via* (13)*.*

XI. Verification, Consolidation, and Continuation Programme

This section consolidates the paper into a self-contained basis for continuation. It (i) records what is *verified* and at what grade of certainty, (ii) states the two open extensions as explicit *conditional* programmes with their single load-bearing hypothesis named, (iii) observes that those two programmes are the same problem, and (iv) documents the experimental protocol that decides the operative regime.

A. Verified status of R1–R4

a) Theorems (unconditional, under stated hypotheses).: theorem VI.1 (R1), theorem VIII.1 and theorem VIII.3 (R4), theorem V.1 (R2), and theorem IV.2/theorem IV.3 *for the diagonal/deep-linear class* (R3). The exponent algebra binding them has been checked symbolically for internal consistency; each of the following is an identity, not a fit:

- $r(\nu) = t(\nu+1)/(1+2t)$ is strictly increasing ($\partial_\nu r = t/(1+2t) > 0$) and crosses $\frac{1}{2}$ exactly at $\nu = 1/(2t)$ (theorem V.1);
- at the worst-case profile $t = s/d^*$, the matching $1/L = 1/(2t) = d^*/(2s)$ gives $L = 2s/d^* = b_0$ (theorem IV.3);
- $\beta(\frac{1}{2}) = 2s/(2s+d^*) = \beta_0$ (eq. (13));
- $\alpha - \beta_0 = 4s^2/[d^*(2s+d^*)] > 0$ for all $s, d^* > 0$ (theorem VIII.3);
- $\alpha/\beta_0 = (2s+d^*)/d^*$, fixing the knee $D_{\text{cross}} = N^{(2s+d^*)/d^*}$ (eq. (16)).

b) Internal-consistency caveat (recorded, not patched).: The empirical anchor $\alpha \approx 0.34$ forces $s/d^* \approx 0.17$, hence $b_0 \approx 0.34 < 1$, while the effective-dimension integral underlying eq. (13) converges only for $b_0 > 1$. The regime that *fits* α therefore lies outside the regime in which the β_0 derivation is valid. This is not a defect to be removed; it is the central empirical message (section X, point 3) and must be carried forward intact by any continuation.

c) Programme (conditional / open).: The nonlinear extension of theorem IV.3 (section XI-B) and the non-perturbative control of Γ_k (section XI-C) are *not* theorems. They are stated below with their single hypothesis made explicit, so that a reader can see precisely what remains to be earned.

B. Programme I: nonlinear μP via a named alignment hypothesis

The obstruction is one of type, not effort. theorem IV.2 is a *variational* (implicit-bias) statement, whereas Tensor Programs IV [6] and the kernel-evolution field theory [9] are *dynamical*: they describe how $K^{(\ell)}(t)$ evolves, not that the stationary measure minimises a norm. No variational principle for the nonlinear μP stationary measure is known, so the Schatten penalty cannot be transported verbatim. What *can* be transported is the exponent's combinatorial origin, once one isolates the single thing the nonlinearity could break.

Assumption XI.1 (Spectral alignment / bounded eigenbasis rotation)**.** *Along training the learned kernel $K^{(L)}(t)$ remains co-diagonalisable with the Laplace–Beltrami basis $\{\psi_k\}$ up to a rotation with summable generator,* $\sup_t \sum_k \left\|\partial_t \hat{\psi}_k(t)\right\|^2 < \infty$*; equivalently the off-diagonal coupling of* (6) *satisfies* $\sup_t \sum_k |\Gamma_k(t)| < \infty$.

Proposition XI.2 (Conditional nonlinear H1)**.** *Adopt depth-μP [7], so that the $L \to \infty$ limit is well defined and the residual-branch scaling makes it commute with $n \to \infty$, and assume theorem XI.1. Then the per-mode gain $g_k(t)$, with $\lambda_k = g_k^2$, evolves as a depth-L multiplicative product whose rich-regime gradient flow selects the Schatten-$2/L$ minimiser modewise; the nonlinearity enters only through a Stein–Price renormalisation of the prefactor μ in* (8) *and a fluctuation term, neither of which alters the exponent. Hence the capacity penalty is $\sum_k \lambda_k^\nu$ with $\nu = 1/L$ and theorem IV.3 extends to nonlinear μP.*

Remark XI.3 (Why the exponent survives but the constant need not)**.** *The Schatten exponent counts the number of multiplicative layer factors in g_k; the chain rule preserves that count for any smooth ϕ, so $\nu = 1/L$ is an architectural invariant while μ absorbs all nonlinear corrections. The endpoints are independently anchored: the large-depth implicit bias of nonlinear networks toward low representation rank (Schatten-0) [8] matches $\nu = 1/L \to 0$. The finite-L interpolation is what theorem XI.2 asserts* under theorem XI.1*; drop that hypothesis and the eigenbasis rotates, the modewise product structure fails, and no clean exponent exists. theorem XI.1 is therefore the entire content of the open problem—the single place where a tautology could re-enter, which is why it is isolated rather than assumed silently.*

C. Programme II: non-perturbative Γ_k via DMFT

Replace the perturbative bound $|\Gamma_k| = O(k^{-\gamma})$ by the Martin–Siggia–Rose/De Dominicis–Janssen generating functional for (6). Introducing a response field $\hat{\lambda}_k$ and integrating out the modes $j \neq k$ yields an effective single-mode action with memory kernel $M(t, t')$ and self-consistent coloured noise ξ_k:

$$\Gamma_k(t) \;=\; \int_0^t M(t,t')\,\lambda_k(t')\,dt' \;+\; \xi_k(t), \qquad \langle \xi_k(t)\xi_k(t')\rangle = C(t,t'), \tag{18}$$

with (M, C) fixed self-consistently by the response and correlation of the remaining modes [9]. Comparing the coupling to the diagonal restoring curvature $2\bar{a}_k^2/(\lambda_k^*)^3$ classifies the dynamics:

- **Weak coupling** ($\|M\| \ll$ curvature): memory negligible, attractor locally stable, $r \to \frac{1}{2}$. This recovers theorem V.2 and the perturbative regime as a limit, and verifies R2.
- **Strong coupling**: the memory kernel drives a transient overshoot $\lambda_k > \lambda_k^*$, during which the operative kernel realises a transient $r > \frac{1}{2}$ and crosses β_0. The control parameter is the ratio of alignment time (set by M) to relaxation time (set by the curvature).

Architecturally, $\|M\| \propto \eta \times$ (coupling strength), the noise amplitude obeys $C \propto 1/(BD)$ from finite-batch/finite-sample fluctuations of $\hat{\Sigma}_D$, and the schedule enters through the time-dependence of V. A rigorous closure of (18)—beyond numerical self-consistency—remains open, as it does for DMFT in this setting generally. A rigorous closure in the weak-coupling phase—with the resulting Adam alignment window and a tight entropy-production frontier—is carried out in section XV; the strong-coupling closure remains the open residue.

D. The two programmes are one problem

theorem XI.1 *is* a statement about Γ_k: bounded eigenbasis rotation is precisely $\sup_t \sum_k |\Gamma_k(t)| < \infty$. Programme II therefore supplies exactly the hypothesis Programme I consumes. The dependency chain for continuation is

$$\begin{aligned}&\underbrace{\text{DMFT closure of } \Gamma_k}_{\text{Programme II}} \Rightarrow \underbrace{\text{eigenbasis-stability bound (theorem XI.1)}}_{\text{the bridge}} \\ &\qquad \Rightarrow \underbrace{\text{nonlinear H1 (theorem XI.2)}}_{\text{Programme I}} \Rightarrow \underbrace{\text{empirical check (section XI-E)}}_{\text{decisive test}}\end{aligned} \tag{19}$$

This ordering never assumes the spectral stationarity it aims to establish; it *derives* the alignment bound dynamically and only then invokes it. That is the structural reason the programme does not reintroduce the tautology of the earlier draft.

E. Verified decisive-test protocol

The test of theorem X.1 has been implemented and its inference logic validated. On synthetic spectra with planted (b, r) the estimator recovers both exponents to within fit-window error; on a teacher–student kernel regression with a border-$\mathcal{H}^s$ teacher it returns $\hat{r}$ with confidence interval covering $\frac{1}{2}$. The verdict rule

$$\hat{r} \approx \tfrac{1}{2} \text{ and flat in } N \;\Rightarrow\; \text{stationary } (\beta = \beta_0), \qquad \hat{r} > \tfrac{1}{2} \text{ and rising in } N \;\Rightarrow\; \text{transient } (\beta = \beta(\hat{r}) \text{ via (13)}),$$

correctly separates planted stationary and transient sweeps across $N \in [10^7, 10^9]$. Two preconditions are mandatory for a trustworthy verdict on trained models. (a) The source exponent must be read from a fit window stable to its endpoints: the empirical kernel carries a finite-size head and an undersampling tail, both of which must be excluded. (b) D must exceed the knee (16); below it, spectral undersampling mimics small r and produces a spurious "stationary" reading. Verdicts are issued on confidence intervals, never point estimates.

F. Further open directions

Beyond (19): (a) deriving d^*_{loc} for concrete compositional target classes, connecting the staircase plateaus [20] to the smooth power law; (b) extending the squared-loss analysis to cross-entropy and to discrete (non-manifold) data, without which the contact with real language models remains qualitative. Both sit down-stream of (19) and do not bear on the status of R1–R4.

XII. Beyond the Barrier: a Spectral Theory of Effective Dimension

The synthesis (section X) localises every genuine feature-learning gain to one of two exits from stationary Sobolev learning: a reduced effective dimension $d^*_{\mathrm{loc}} < d^*$, or transient kernel alignment. Section XI-C addresses the second exit. This section addresses the first. It is the programme named, but deferred, in section XI-F: *deriving d^*_{loc} for concrete target classes and connecting it to the staircase plateaus of [20].*

I keep the parent paper's discipline. The earlier draft's "sublinear mode count" $K_{\mathrm{eff}} \asymp N^{2s/(2s+d^*)}$ was an *ad hoc* postulate, and theorem VIII.1 showed it is suboptimal on a Sobolev ball. The aim here is the opposite of postulation: to *derive* K_{eff}, and d^*_{loc}, from named properties of the target, and to be explicit about which structures buy a genuine exponent and which merely close the gap between feature learning and the lazy limit. I will find that the two structures the parent paper bundled together under "$d^*_{\mathrm{loc}} < d^*$" (section VIII) are not the same kind of object, and separating them sharpens rather than weakens the regime statement.

A. The effective dimension, defined

Write the best N-term (adaptive) approximation error of the target in the operative dictionary as $\sigma_N(f^*)^2_{L^2}$, and let $\{\bar{a}_{(k)}\}_{k\geq 1}$ denote the coefficients $\{\bar{a}_k\}$ of (3) rearranged into non-increasing magnitude. I separate two source exponents of the same target, exactly as section II separated two kernels.

Definition XII.1 (Linear and adaptive source exponents)**.** *The* linear *source exponent t is the unsorted decay of* (3)*, $\bar{a}_k^2 \asymp k^{-(1+2t)}$. The* adaptive *source exponent $\tilde{t}$ is the sorted decay,*

$$\bar{a}^2_{(k)} \asymp k^{-(1+2\tilde{t})}, \qquad \tilde{t} \geq t, \tag{20}$$

the inequality because rearrangement cannot slow the tail. The effective dimension *is*

$$\boxed{d^*_{\mathrm{loc}} := \frac{s}{\tilde{t}}} \qquad \big(\leq d^* = s/t\big), \tag{21}$$

so that the attainable (feature-learned) approximation rate is $\mathcal{E}_{\mathrm{appr}}(N) \asymp N^{-\alpha^\star}$ with $\alpha^\star = 2s/d^*_{\mathrm{loc}} = 2\tilde{t}$, while the lazy / ordered-basis rate is fixed at $\alpha = 2s/d^* = 2t$ as in theorem VIII.1.

Two immediate sanity checks anchor the definition.

Lemma XII.2 (Sobolev recovers $d^*_{\mathrm{loc}} = d^*$)**.** *If f^* ranges over the Sobolev ball $\{\|f\|_{\mathcal{H}^s} \leq R\}$ then, in the worst case over the ball, $\tilde{t} = t = s/d^*$ and $d^*_{\mathrm{loc}} = d^*$; adaptivity buys no exponent.*

Proof. The extremal element of the ball spreads its $\mathcal{H}^s$-budget uniformly across a dyadic shell: $\bar{a}_k^2 \asymp k^{-(1+2s/d^*)}$ already non-increasing, so sorting is the identity and $\tilde{t} = t$. This is theorem VIII.1 read through (21): the Laplacian eigenbasis is rate-optimal for $\mathcal{H}^s$, so the best N-term and linear widths coincide. □

Remark XII.3 (Why a single scalar suffices)**.** *Both estimation and approximation in this paper collapse onto a single source exponent (here $r = \frac{1}{2}$ in theorem V.2; there $\tilde{t}$). The content of d^*_{loc} is therefore not a new mechanism bolted on, but the statement that the* same *spectral bookkeeping that pinned stationary learning to β_0 also controls the only quantity that can move the approximation exponent. What changes the rate is what changes $\tilde{t}$.*

B. Inhomogeneous targets: Besov sparsity closes the adaptivity gap

The first structure is spatial inhomogeneity. Let $f^* \in B^s_{p,q}(\mathcal{M})$ with $p < 2$, characterised in the eigenbasis by the usual Besov sequence norm, and write the *sparsity defect*

$$\delta := \tfrac{1}{p} - \tfrac{1}{2} > 0. \tag{22}$$

Proposition XII.4 (Adaptivity gap and the derived K_{eff})**.** *Assume the supercritical regime $s/d^* > \delta$ (equivalently $f^* \in L^2$ with nonlinear approximation admissible). Then, by the linear/nonlinear width dichotomy for Besov balls [15], [16],*

$$\underbrace{\mathcal{E}^{\text{lazy}}_{\text{appr}}(N) \asymp N^{-2(s/d^*-\delta)}}_{\textit{ordered basis}} \;<\; \underbrace{\mathcal{E}^{\text{adapt}}_{\text{appr}}(N) \asymp N^{-2s/d^*}}_{\textit{best } N\textit{-term}}, \tag{23}$$

i.e. feature learning recovers an exponent gain of exactly $2\delta = 2(1/p - 1/2)$ over the lazy limit. Consequently, to reach a fixed squared-error budget the adaptive method resolves

$$K_{\text{eff}}(N) \;\asymp\; N^{1-d^*\delta/s} \;\ll\; N, \qquad \textit{exponent } \theta_{\text{Besov}} = 1 - \frac{d^*}{s}\Big(\frac{1}{p} - \frac{1}{2}\Big) \in (0,1), \tag{24}$$

ordered modes worth of the lazy budget N. At $p = 2$ ($\delta = 0$) this is $K_{\text{eff}} = N$ and (23) *collapses to theorem VIII.1.*

Proof sketch. For $p < 2$ the Kolmogorov (linear) N-width of the Besov ball in L^2 loses δ orders of smoothness through the embedding $B^s_{p,q} \hookrightarrow \mathcal{H}^{s-d^*\delta}$, giving the linear rate exponent $s/d^* - \delta$; the best N-term width retains the full s/d^* above the embedding line [16]. Squaring gives (23). For K_{eff}, fix budget ε: the lazy method needs $N \asymp \varepsilon^{-d^*/(s-d^*\delta)}$ ordered modes, the adaptive method $K_{\text{eff}} \asymp \varepsilon^{-d^*/s}$ adaptively placed ones; eliminating ε yields (24). □

Remark XII.5 (Besov buys an *adaptivity* gain, not an exponent below $2s/d^*$)**.** *This is the point at which I must depart from a tempting but false unification. Measured* per adaptively chosen parameter, *the Besov rate is still $2s/d^*$: a depth-μP network spending N parameters adaptively reaches N^{-2s/d^*}, the same exponent as on a Sobolev target. Sparsity does not push the parameter-rate below $2s/d^*$; in worst case over the ball the sorted decay is still $\tilde{t} = s/d^*$, so by* (21) *$d^*_{\text{loc}} = d^*$. What sparsity does is make the* lazy *method strictly suboptimal, so that feature learning has something to recover — precisely the 2δ gap. This is the regime where [11] report feature learning improving the scaling law, and our reading is sharper than the parent paper's section VIII bullet: Besov $p < 2$ is the home of the* feature-learning-beats-lazy *phenomenon, not of a sub-Sobolev exponent. The latter requires composition.*

C. Compositional targets: the genuine exponent reduction

The second structure is hierarchy, and it is the only one that moves the parameter-rate. Let

$$f^* = g_L \circ g_{L-1} \circ \cdots \circ g_1, \tag{25}$$

where each constituent g_i is $\mathcal{H}^s$-smooth and depends on a subset of coordinates (or an intermediate representation) of intrinsic dimension $d_i \ll d^*$.

Theorem XII.6 (Compositional rate and $d^*_{\text{loc}} = \max_i d_i$)**.** *Let a depth-$L$ network whose connectivity realises the composition DAG of* (25) *approximate f^*. Then*

$$\mathcal{E}_{\text{appr}}(N) \asymp N^{-2s/d^*_{\text{loc}}}, \qquad \boxed{d^*_{\text{loc}} = \max_{1\le i\le L} d_i}, \qquad \alpha_{\text{comp}} = \frac{2s}{d^*_{\text{loc}}} \;\ge\; \frac{2s}{d^*} = \alpha, \tag{26}$$

with equality iff some constituent is full-dimensional. Equivalently the sorted source exponent is $\tilde{t} = s/\max_i d_i > t$, consistent with (21).

Proof sketch. Approximating each constituent g_i to accuracy ε in L^∞ costs $n_i \asymp \varepsilon^{-d_i/s}$ units, the Sobolev rate *in its own* dimension d_i. Because the constituents are smooth ($\phi'' \in L^\infty$ as in (4)), errors propagate through the composition with bounded amplification: $\left\|\hat{f} - f^*\right\|_\infty \lesssim \sum_i \mathrm{Lip}(g_{>i})\,\varepsilon_i$, a finite ($L$-term, $L = O(1)$ in N) sum [16], [19]. Allocating accuracy so that every constituent contributes equally, the total unit count is dominated by the hardest constituent, $N \asymp \max_i \varepsilon^{-d_i/s} = \varepsilon^{-\max_i d_i/s}$. Inverting gives $\varepsilon \asymp N^{-s/\max_i d_i}$ and (26) on squaring. The shallow network, unable to realise the DAG, sees only the ambient d^* and is held to $\alpha = 2s/d^*$ — the representational separation of [19]. □

Remark XII.7 (Contact with the saddle-to-saddle staircase)**.** *Theorem XII.6 is a statement about the* rate*; [20] is a statement about the* dynamics *of reaching it. They meet as follows. SGD on a hierarchical target learns the constituents in order of increasing leap complexity, with a saddle plateau between successive levels. Identify the leap at level i with its local dimension d_i: the network sits on a plateau while the loss is bottlenecked by an as-yet-unresolved constituent, then drops when that constituent is learned. The asymptotic exponent $2s/\max_i d_i$ is set by the* last *(hardest, highest-d_i) level to be resolved; the plateaus are the pre-asymptotic transients on the way there. This is a derivation of the qualitative connection section XI-F left open; the leap = local-dimension identification, previously stated as a hypothesis, is itself derived in theorem XII.8 below via a per-level Freidlin–Wentzell analysis, modulo only the transfer of the saddle-to-saddle structure of [20] to the manifold setting.*

Theorem XII.8 (Per-level metastability derives (H-leap))**.** *Adopt the saddle-to-saddle structure of [20] for the spectral flow* (31) *at small noise $D_k = \eta T \to 0$, with rate functional* (33)*. Let plateau ℓ be the metastable spectrum at which constituents $1, \dots, \ell - 1$ are resolved and $g_\ell, \dots, g_L$ are not. Then:*

(i) *plateau ℓ is a hyperbolic saddle of V_{tot} whose unstable manifold is the d_ℓ-dimensional mode subspace of constituent ℓ; the resolved constituents sit at stable values and, being at their own minima, contribute no least-action descent, so the Freidlin–Wentzell escape path lives in the constituent-ℓ subspace alone;*

(ii) *the plateau exit time obeys the Kramers/Eyring law [24], [35]*

$$\tau_\ell \asymp \exp\bigl(\Delta V_\ell / T\bigr), \qquad \Delta V_\ell \asymp d_\ell \cdot v_*, \tag{27}$$

where v_* is the per-direction activation barrier (the energy to seed one effective dimension of g_ℓ above the noise floor) and the simultaneously activated *directions number $\asymp d_\ell$ (the leap), by Weyl counting of the constituent's modes; the barrier depends on the* local *d_ℓ, never on the ambient d^*;*

(iii) *consequently the leap complexity of [20] equals d_ℓ/s (resolution decades of a d_ℓ-dimensional Sobolev constituent), the constituents are learned in order of increasing d_ℓ (smaller $d_\ell \Rightarrow$ lower barrier $\Rightarrow$ earlier escape), and the asymptotic exponent is set by the last/hardest level* $\max_i d_i = d^*_{\mathrm{loc}}$*, recovering theorem XII.6.*

This is the identification (H-leap) *of theorem XII.7, now derived rather than assumed, modulo the transfer of the saddle-to-saddle structure of [20] from hypercube SGD to the manifold/Langevin setting (theorem XII.9).*

Remark XII.9 (What is proved and the one transfer that remains)**.** *Given the saddle structure, parts* (i)–(iii) *are theorem-grade: the restriction of the Freidlin–Wentzell action to the unstable manifold is exact for a hyperbolic saddle with frozen spectators, and the sharp Kramers prefactor and exponent are [35]. The barrier scaling $\Delta V_\ell \asymp d_\ell\, v_*$ is a Weyl-counting argument at the grade of the approximation results of section XII. What is* not *reproved here is that* (31) *on a manifold inherits the exact saddle-to-saddle plateau sequence that [20] establishes for SGD on the hypercube.*

This transfer is a named residual hypothesis*: the geometry of the hypercube (product structure, discrete symmetry) is used in [20] to isolate the saddle plateaux, and transporting that isolation to Langevin on a Riemannian manifold would require showing that the Morse-theoretic saddle classification of V_{tot} is preserved under the change of dynamics. No such transfer is currently in the literature, and I do not claim it.*

The residual hypothesis is, however, mild in the following sense: it bears only *on the microscopic saddle-to-saddle dynamics—specifically on which constituents are learned in which order and on the leading constant of the plateau exit time—not on the* asymptotic rate. *The rate* $2s/d^*_{\mathrm{loc}}$ *of theorem XII.6 holds regardless: it is a minimax statement about representation that does not reference dynamics at all. The residual therefore does not touch R1, R2, R3, or R4; it affects only the pre-asymptotic staircase structure of theorem XII.7 and theorem XII.8(ii), and only through the precise constant* v_* *in* (27)*, not through the exponent* d_ℓ *itself.*

D. Heuristic contact with language

I record, at the explicit grade of *heuristic* (cf. the discrete-data caveat of section XI-F), how (25) reads for language. A plausible hierarchy places local-n-gram statistics at small d_i (a few tokens of context), syntactic structure at intermediate d_i (dependency arcs over a clause), and long-range semantic / discourse structure at the largest d_i. The prediction of theorem XII.6 is then that the *asymptotic* exponent is set by the semantic level (the largest d_i), while the visible learning curve passes through plateaus as cheaper levels saturate — the empirical phase-transition phenomenology. I stress that this is contact, not contact at the rate of theorem: squared loss is not cross-entropy, and discrete tokens are not a manifold; both gaps are exactly those flagged in section XI-F and neither is closed here.

E. The multi-level regime map

Section IX split the $(\log N, \log D)$ plane with the single knee $D_{\mathrm{cross}} = N^{(2s+d^*)/d^*}$. With structure that knee fans out into a pencil of lines, one per level. Model the target as an additive mixture across complexity scales, $f^* = \sum_i w_i\,\phi_i$, where ϕ_i is a structure of local dimension d_i carrying energy w_i^2 (the compositional case behaves the same pre-asymptotically, the additive case being the tractable surrogate). Level i contributes, when binding,

$$\text{approximation } N^{-2s/d_i} \qquad \text{and} \qquad \text{estimation } D^{-\beta_0(d_i)}, \quad \beta_0(d_i) = \frac{2s}{2s+d_i}, \tag{28}$$

each a copy of R1/R4 *at the level's own dimension*. The two errors of level i trade off along that level's own crossover line

$$\boxed{D = N^{(2s+d_i)/d_i} = N^{\,1+2s/d_i}}, \tag{29}$$

whose slope $1 + 2s/d_i$ is *steeper for smaller* d_i.

Proposition XII.10 (Stratification of the plane)**.** *Order the levels* $d_1 < d_2 < \cdots < d_L$*. The lines* (29) *are nested by slope and foliate the first quadrant into* $L{+}1$ *cells. In the cell between the level-*i *and level-*$(i{+}1)$ *knees, the binding (largest-residual) contribution is the approximation error of level* $i{+}1$ *on the data-rich side and the estimation error of level* i *on the data-poor side: low-dimensional structure is resolved first (cheap in both* N *and* D*), and the operative bottleneck climbs the hierarchy as either resource grows. The compute-optimal allocation* $C = ND$ *therefore tilts more data-heavily the higher the binding level, since* $\alpha - \beta_0 = 4s^2/[d_i(2s + d_i)]$ *(theorem VIII.3)* grows *as* d_i *shrinks.*

Sketch. Each level is an independent R1/R4 problem at dimension d_i, so its kink is (29) by the algebra of (16). The binding level is the one with the slowest-decaying residual at (N, D); as N, D increase along any ray, residuals vanish in increasing-d_i order because the rate exponents $2s/d_i$ and $\beta_0(d_i)$ are both decreasing in d_i. The allocation statement is theorem VIII.3 read per level. □

Remark XII.11 (Two staircases, and how to tell them apart)**.** *Theorem XII.10 predicts kinks in the learning curve from* resource *crossings (a level's approximation and estimation errors swap dominance); the saddle-to-saddle picture of theorem XII.7 predicts kinks from* dynamics *(a constituent is learned). An observed staircase in a trained LLM could be either. They are distinguishable by the decisive test of theorem X.1: a resource crossing is a static feature of the loss surface and leaves the operative source exponent at* $r = \frac{1}{2}$

(stationary, β_0 at the binding dimension), whereas a dynamical saddle escape is a transient and shows r rising as the level is resolved. The parent paper's verdict rule thus extends verbatim to the multi-level setting — it now reads off which *staircase one is on, not merely stationary-versus-transient.*

F. Optimal depth as hierarchy navigation

Theorem IV.3 identified the load-bearing depth $L = 2s/d^* = b_0$ as the depth–smoothness matching at which the stationary attractor saturates the barrier: the implicit Schatten exponent $\nu = 1/L$ (theorem IV.2) meets the target's intrinsic smoothness scale $1/(2t) = d^*/(2s)$. The same matching, run at the effective dimension, gives the structured-data analogue.

Theorem XII.12 (Matched depth at effective dimension)**.** *On a target with effective dimension d^*_{loc} and adaptive source exponent $\tilde{t} = s/d^*_{\mathrm{loc}}$, the depth at which the stationary attractor saturates the d^*_{loc}-*dimensional *barrier is*

$$\frac{1}{L^\star} = \frac{1}{2\tilde{t}} = \frac{d^*_{\mathrm{loc}}}{2s} \iff \boxed{L^\star = \frac{2s}{d^*_{\mathrm{loc}}}} \geq \frac{2s}{d^*}, \tag{30}$$

so more structured data (smaller d^*_{loc}) demands a deeper *matched network.*

Proof. Repeat theorem IV.3 with the source exponent of the effective problem. The attractor (10) on the operative kernel realises $r(\nu) = \tilde{t}(\nu+1)/(1+2\tilde{t})$ (theorem V.1 with $t \mapsto \tilde{t}$); the saturation $r = \frac{1}{2}$ occurs at $\nu_\star = 1/(2\tilde{t})$, and $\nu = 1/L$ gives (30). Monotonicity of $L^\star$ in $1/d^*_{\mathrm{loc}}$ is immediate. □

Remark XII.13 (Representational vs. statistical depth)**.** *Composition imposes* two *lower bounds on depth, and they are different objects. theorem XII.6 needs $L \geq L_{\mathrm{hier}}$, the number of compositional levels, simply to* realise *the rate $2s/d^*_{\mathrm{loc}}$ (a representational requirement, [19]). theorem XII.12 needs $L = L^\star = 2s/d^*_{\mathrm{loc}}$ to* saturate the barrier *statistically at that rate. The two coincide only by accident; in general the operative depth must satisfy both, $L \geq \max\{L_{\mathrm{hier}}, L^\star\}$. The realisability caveat of theorem IV.4 carries over: $L^\star = 2s/d^*_{\mathrm{loc}}$ is an integer only when $s \geq d^*_{\mathrm{loc}}$, so a sufficiently rough constituent ($s \ll d_i$) cannot be matched by any finite-depth stationary network — the same boundary that foreshadowed section X, now read per level.*

The empirical reading is that architectures should deepen as data becomes more compositional. This is consistent with the observed utility of depth on structured / reasoning-heavy tasks, and it is a prediction with a sign: holding s fixed, the matched depth scales as $1/d^*_{\mathrm{loc}}$, so a task whose hardest constituent halves in dimension should reward roughly a doubling of matched depth.

G. Status of section XII

In the grading discipline of section XI-A:

a) Theorems (under stated hypotheses).: theorem XII.2 (Sobolev gives $d^*_{\mathrm{loc}} = d^*$, a re-reading of theorem VIII.1); theorem XII.4 (the 2δ adaptivity gap and the derived $K_{\mathrm{eff}} \asymp N^{1-d^*\delta/s}$, from [16]); theorem XII.6 (the compositional rate $2s/\max_i d_i$, from [19]); theorem XII.10 (stratification, by per-level R1/R4 algebra); theorem XII.12 (matched depth $L^\star = 2s/d^*_{\mathrm{loc}}$, from theorems IV.3 and V.1). The exponent identities — $\theta_{\mathrm{Besov}} = 1$ at $p = 2$, $\alpha_{\mathrm{comp}} = 2s/d^*_{\mathrm{loc}}$, $L^\star = 2s/d^*_{\mathrm{loc}}$, the per-level knee (29) — reduce to the parent paper's identities when $d^*_{\mathrm{loc}} = d^*$, and have been checked for that consistency.

b) The sharpened regime statement (a correction, not a postulate).: The parent paper's section VIII bundled Besov and compositional structure as "$d^*_{\mathrm{loc}} < d^*$." theorem XII.5 separates them: Besov $p < 2$ closes the *lazy-to-adaptive* gap but leaves the parameter-rate at $2s/d^*$ (so $d^*_{\mathrm{loc}} = d^*$ in worst case); only composition delivers $d^*_{\mathrm{loc}} < d^*$ and a sub-Sobolev exponent. This refines the parent claim and removes the one place where "d^*_{loc}" could have been read as a free knob.

c) Hypotheses (named, open).: Two identifications are used but not proved, and are flagged so a reader sees exactly what is owed. **(H-leap)** the leap complexity of [20] equals the local dimension d_i of the constituent it resolves (theorem XII.7); this is now *derived* in theorem XII.8 (per-level Kramers law, barrier $\asymp d_i$), leaving only the transfer of the saddle-to-saddle structure of [20] to the manifold/Langevin setting as a milder residual hypothesis (theorem XII.9). **(H-lang)** the language hierarchy of section XII-D has the assumed level/dimension structure; this is heuristic and inherits the cross-entropy and discreteness gaps of section XI-F. Neither hypothesis bears on R1–R4 or on the theorems above; both sit strictly downstream, in the $d^*_{\text{loc}} < d^*$ exit whose existence — not whose microscopics — is all the synthesis of section X requires.

d) Decisive tests.: The protocol of theorem X.1 extends to two new readings. *(i) Effective dimension:* fit the binding-level knee slope $1+2s/d_i$ from the loss surface (eq. (29)) and the approximation exponent $\alpha = 2s/d^*_{\text{loc}}$ from the N-sweep; consistency of the two recovered d^*_{loc} values certifies the effective-dimension picture and, by contrast with d^* measured from the data manifold, quantifies the compositional gain. *(ii) Staircase type:* apply theorem XII.11 — r flat at $\frac{1}{2}$ across a kink marks a resource crossing (theorem XII.10); r rising marks a dynamical saddle escape (theorem XII.7).

XIII. Beyond the Barrier II: Non-equilibrium Spectral Dynamics and Driven Optimisers

Programme II (section XI-C) already isolated the branch in which the barrier is evaded: the *strong-coupling* regime of (18), where the memory kernel drives a transient overshoot and the operative kernel realises $r > \frac{1}{2}$. Section XII pursued the *first* exit—lowering d^*_{loc}, a property of the target. This section develops the *second* exit—the non-equilibrium dynamics that hold $r > \frac{1}{2}$ *during* training—to the same standard: separating what is a theorem from a controlled heuristic and a measurable prediction. The thesis I defend, and which *sharpens* rather than overturns R1, is a single sentence: $r > \frac{1}{2}$ *is sustainable only as a driven non-equilibrium steady state, paid for by a strictly positive entropy-production rate; remove the drive and the spectrum relaxes to the critical attractor* $\boldsymbol{\lambda}^*$ *of theorem V.2.* Driving therefore governs the pre-asymptotic prefactor and the time-to-asymptote, not the worst-case asymptotic exponent, which remains pinned by R1 unless the target also supplies structural $d^*_{\text{loc}} < d^*$ (section XII).

A. The spectral flow as a stochastic process

With finite batch B and finite sample D, the empirical $\hat{\Sigma}_D$ in (5) fluctuates; projecting onto the modes $\{\psi_k\}$ promotes the spectral system (6) to a Langevin equation

$$d\lambda_k = \left[-\eta\,\frac{\partial V}{\partial \lambda_k} + \Gamma_k\right]dt + \sqrt{2D_k}\,dW_k, \tag{31}$$

with V from (8), Γ_k the inter-mode coupling of (18), and per-mode diffusion D_k set by the mini-batch/sample fluctuations of $\hat{\Sigma}_D$; from the noise of (18), $D_k \propto C/(BD)$ modulated multiplicatively by λ_k. I state (31) as a modelling step: freezing the coloured self-consistent noise of (18) to a white, leading-order D_k is a heuristic, exact only in the weak-coupling limit.

a) The pivotal dichotomy: equilibrium versus non-equilibrium.: Write $V_{\text{tot}} = V + \Phi$ with $\Gamma_k = -\eta\,\partial\Phi/\partial\lambda_k$ (the gradient-coupling hypothesis of section V). The Fokker–Planck equation for the spectral measure $P(\boldsymbol{\lambda}, t)$ is

$$\partial_t P = \sum_k \partial_{\lambda_k}\Big[\eta\,(\partial_{\lambda_k} V_{\text{tot}})\,P + \partial_{\lambda_k}\big(D_k P\big)\Big]. \tag{32}$$

Proposition XIII.1 (Equilibrium relaxation recovers R2)**.** *If* (i) *the drift is the gradient of* V_{tot} *and* (ii) *a fluctuation–dissipation relation* $D_k = \eta T$ *holds with a single,* $\boldsymbol{\lambda}$*-independent temperature* T*, then* (32) *is reversible with the unique stationary density* $P_\infty \propto \exp(-V_{\text{tot}}/T)$*, which concentrates as* $T \to 0$ *on the attractor* $\boldsymbol{\lambda}^*$ *of* (10)*. The probability current* $J_k \equiv 0$ *and the entropy-production rate* (34) *vanishes.*

This is the stochastic-thermodynamic restatement of theorem V.2: the *relaxed* kernel is critical, $r = \frac{1}{2}$. The correction to the naive reading is that its hypotheses are not generic.

Remark XIII.2 (Stochastic gradient descent generically breaks detailed balance)**.** *The minibatch noise covariance is state-dependent and anisotropic—aligned with the per-mode curvature—so D_k is not a common ηT, and a shared minibatch correlates the noise across modes [22], [23]. Then the stationary solution of the (matrix-diffusion) generalisation of* (32) *carries a non-vanishing current $J_k \neq 0$: a genuine* non-equilibrium steady state *(NESS), even at "convergence". The long-time operative kernel need not be $\boldsymbol{\lambda}^*$; it can sustain a current-carrying spectrum with $r \neq \frac{1}{2}$. This does not contradict theorem V.2, which concerns the* equilibrium *attractor; it identifies the exact assumption—detailed balance / FDR—under which "stationary" coincides with "critical". Everything below is a controlled way of breaking that assumption.*

B. Action functional and the saddle-to-saddle picture

For (31) the path measure obeys, as $D_k \to 0$, a large-deviation principle with Onsager–Machlup/Freidlin–Wentzell rate functional [24], [25]

$$S[\boldsymbol{\lambda}(\cdot)] = \frac{1}{4}\int_0^\tau \sum_k \frac{1}{D_k}\Big(\dot{\lambda}_k + \eta\,\partial_{\lambda_k} V_{\mathrm{tot}} - \Gamma_k^{\mathrm{nc}}\Big)^2 dt, \tag{33}$$

where Γ_k^{nc} is the non-conservative part of the coupling. Minimisers are the instantons of the spectral flow. Between two metastable spectra—e.g. two plateaux of the staircase, theorem XII.11—the least-action path is a hopping instanton whose duration is set by the barrier of V_{tot} along the activating mode; a time-ordered sequence of such hops is precisely the *saddle-to-saddle* (dynamical-staircase) dynamics. The weak/strong-coupling classification of (18) reappears as the large/small-action regime. Status: the LDP is rigorous for a fixed finite mode truncation with white noise (Freidlin–Wentzell); the infinite-mode, coloured-noise closure is the same open DMFT step flagged in section XI-C.

C. Entropy production and a log-only non-stationarity criterion

Definition XIII.3 (Spectral entropy-production rate)**.** *For the (generalised) NESS of* (32) *with current $J_k = -[\eta\,(\partial_{\lambda_k} V_{\mathrm{tot}})P + \partial_{\lambda_k}(D_k P)]$ plus any non-conservative drift, set*

$$\sigma = \sum_k \frac{J_k^2}{D_k\,P} \;\geq\; 0, \tag{34}$$

with $\sigma = 0$ iff every $J_k = 0$ iff detailed balance holds.

Proposition XIII.4 (Non-stationarity costs dissipation)**.** *In steady state $r(t)$ departs from $\frac{1}{2}$ only if $\sigma > 0$; equivalently, the critical spectrum $\boldsymbol{\lambda}^*$ ($r = \frac{1}{2}$) is the unique zero-entropy-production steady state. Holding $r > \frac{1}{2}$ is thus a driven process with a strictly positive power budget $\dot{W} = T\sigma$.*

The useful corollary is a *measurable* surrogate that needs only training logs, not weights. Along training one can snapshot the empirical-NTK spectrum $\hat{\lambda}_k(t)$ with the decisive-test instrumentation of section XI-E.

Remark XIII.5 (A criterion from logs)**.** *Two log-only observables track non-stationarity:* (i) *the normalised kernel velocity $\left\|\dot{K}\right\|_F / \|K\|_F$ (equivalently $\left\|\dot{\boldsymbol{\lambda}}\right\|$), which vanishes at the attractor and is positive on the driven/transient branch; and* (ii) *the drift $\dot{\hat{r}}(t)$ of the fitted source exponent from the section XI-E estimator. A non-zero, non-decaying $\left\|\dot{K}\right\|_F$ together with $\dot{\hat{r}} > 0$ is the dynamical signature of the strong-coupling branch of* (18)*, separating a genuine transient (r rising) from a critical attractor (r flat). I do not claim $\left\|\dot{K}\right\|_F = \sigma$: it is a one-sided proxy ($\sigma = 0 \Rightarrow \dot{K} \to 0$; the converse needs the FDR of theorem XIII.1). This is the implementable form of "measure non-stationarity without the weights".*

D. Driven optimisers: a steady state with $r > \frac{1}{2}$

I now make the construction explicit, separating two mechanisms that are easy to conflate: one that *moves the steady state*, one that only *reshapes the transient*.

1) (A) Spectral energy injection moves the steady state: Add a forcing $F_k = \varepsilon\, \lambda_k^{-\gamma}$ ($\varepsilon, \gamma > 0$) to (31)—a non-conservative drift that pumps the weak (small-λ, large-k) modes.

Proposition XIII.6 (Driven NESS spectrum)**.** *Using $\partial_{\lambda_k} V = -\bar{a}_k^2/\lambda_k^2 + \mu\nu\lambda_k^{\nu-1}$ from* (8)*, the deterministic steady state $\dot{\lambda}_k = 0$ of* (31)+F *obeys $\eta\bar{a}_k^2\lambda_k^{-2} = \varepsilon\lambda_k^{-\gamma} + \eta\mu\nu\lambda_k^{\nu-1}$. In the tail, where the injection dominates the capacity term ($\gamma < 1-\nu$), the operative balance $\eta\bar{a}_k^2\lambda_k^{-2} \approx \varepsilon\lambda_k^{-\gamma}$ gives*

$$\lambda_k^{\mathrm{drive}} \propto (\bar{a}_k^2)^{1/(2-\gamma)} \asymp k^{-b_{\mathrm{drive}}}, \qquad b_{\mathrm{drive}} = \frac{1+2t}{2-\gamma}, \tag{35}$$

and, since the source-exponent computation of theorem V.1 depends on the spectrum only through b,

$$r_{\mathrm{drive}} = \frac{t}{b_{\mathrm{drive}}} = \frac{t(2-\gamma)}{1+2t}, \tag{36}$$

strictly increasing in γ, with $r_{\mathrm{drive}} > \frac{1}{2} \iff \gamma < 1 - \frac{1}{2t}$ (so injection lifts r past $\frac{1}{2}$ only for $t > \frac{1}{2}$; for rougher free targets it raises r toward, but not beyond, the critical value within the stable range).

Theorem XIII.7 (Stability ceiling; no free lunch)**.** *The driven spectrum* (35) *is a steady state of* (31)+F only while F is applied*: setting $\varepsilon = 0$ returns the flow to $\boldsymbol{\lambda}^*$ of* (10) *by the LaSalle argument of section V, since F is the only non-gradient term. Discrete-time stability requires the per-mode effective step obey a descent-lemma bound $\eta_k\,\kappa_k \le 2$ with head curvature $\kappa_{\mathrm{head}} \asymp 2\bar{a}_{\mathrm{head}}^2/(\lambda_{\mathrm{head}}^*)^3$; this caps ε and forbids $\gamma \ge 1-\nu$, beyond which injected tail energy is not absorbed and the iteration diverges. Hence* (i) *$r_{\mathrm{drive}} > \frac{1}{2}$ is attainable but only as a driven NESS;* (ii) *the sustainable boost $\Delta r = r_{\mathrm{drive}} - \frac{1}{2}$ is bounded by the spectral dynamic range and the stability ceiling;* (iii) *the state carries $\sigma > 0$ (theorem XIII.3), i.e. a permanent power budget $\dot{W} = T\sigma$.*

Remark XIII.8 (What this does and does not do to R1)**.** *R1 (theorem VI.1) is an information-theoretic lower bound on the* estimation *error from D samples over the worst-case $\mathcal{H}^s$ ball; no drift or noise schedule evades it, because it never references the optimiser. A driven NESS changes the* operative kernel*—hence the constant and the pre-asymptotic exponent realised at finite D—and on targets genuinely easier than worst-case it realises $\beta(r_{\mathrm{drive}}) > \beta_0$ over the accessible range. On the worst-case ball the over-aligned driven kernel is misspecified exactly as in theorem V.2 (ii), and the realised rate reverts to β_0 as $D \to \infty$ once the drive's finite budget is amortised. Driving buys a prefactor and a faster approach to the asymptote; an asymptotic* exponent *gain still requires $d_{\mathrm{loc}}^* < d^*$ (section XII). The two exits are complementary: section XII moves the exponent, section XIII moves the prefactor and the time-to-asymptote, at a quantified thermodynamic cost. This is the precise sense in which R1 is* localised*—to the relaxed, undriven, worst-case asymptote—rather than overturned.*

2) (B) Per-mode learning rate reshapes the transient, not the steady state: Set $\eta_k = \eta_0\lambda_k^{-\delta}$ ($\delta > 0$).

Proposition XIII.9 (A positive preconditioner does not move the attractor)**.** *Because $\eta_k > 0$ multiplies a gradient drift, the fixed-point set $\partial_{\lambda_k} V_{\mathrm{tot}} = 0$ is unchanged: the steady state remains $\boldsymbol{\lambda}^*$. What changes is the* relaxation spectrum*—weak modes equilibrate faster by a factor $\lambda_k^{-\delta}$, inverting the natural timescale ordering. The flow therefore passes through a* transient *overshoot in which the tail is filled before the head relaxes, realising a transient operative $r(t) > \frac{1}{2}$ en route to $\boldsymbol{\lambda}^*$, with overshoot duration scaling as the head/tail timescale separation $(\lambda_{\mathrm{head}}^*/\lambda_{\mathrm{tail}}^*)^\delta$.*

This is the timescale (saddle-to-saddle) mechanism of section XIII-B, and it is *distinct* from injection: it engineers a longer-lived transient, not a new steady state. I flag this as a correction to the natural but mistaken expectation that a per-mode rate "shifts $\boldsymbol{\lambda}^*$ rightward"; it shifts the *approach*, not the fixed point. A genuine steady-state shift requires either a non-conservative force (A) or broken FDR (C).

3) (C) Anisotropic noise engineers the NESS through broken FDR: Inject noise with per-mode covariance $D_k \propto k^{-\zeta}$. By the dichotomy of theorem XIII.2, choosing D_k not proportional to the dissipation breaks FDR, so the stationary density is a NESS with $J_k \neq 0$. To leading order in a small-noise expansion the NESS mean shifts by $\delta\lambda_k \propto (\partial_{\lambda_k} D_k)/\kappa_k$, a ζ-dependent tilt of the spectrum; tuning ζ tilts toward a flatter spectrum (larger r) at the cost $\sigma \propto \sum_k (\partial_{\lambda_k} D_k)^2/(\cdots) > 0$. Status: leading-order heuristic (the full NESS density is not closed-form); the qualitative claim—that anisotropic noise is a control knob for steady-state r, paid in σ—is robust and is the noise-side dual of injection (A).

E. Adam as an implicit driven optimiser

Theorem VII.1 already states that Adam, by rescaling the metric, is not a stationary-kernel method—any gain it produces is, by definition, kernel modification. Here I identify *which* of the mechanisms above it implements.

a) Setup (diagonal-in-eigenbasis approximation).: Adam preconditions by $P_t = \mathrm{diag}(v_t^{-1/2})$ with v_t a running mean of g_t^2 [26]. The gradient with respect to the mode-k coefficient has magnitude $|g_k| \propto \lambda_k\,|\mathrm{res}_k|$ (feature scale $\times$ residual). Under the approximation that parameter coordinates align with the NTK eigenbasis—exact for the diagonal-linear class of theorem IV.2, approximate otherwise, which I flag—one has $v_k \propto \lambda_k^2\,\mathrm{res}_k^2$, so the effective per-mode step is

$$\eta_k^{\mathrm{Adam}} \propto \frac{1}{\sqrt{v_k}} \propto \frac{1}{\lambda_k\,|\mathrm{res}_k|}. \tag{37}$$

Equation (37) reads as two superposed drives. The $1/\lambda_k$ factor is the per-mode rate of mechanism (B) with $\delta = 1$—the strongest tail amplification short of the stability ceiling of theorem XIII.7. The $1/|\mathrm{res}_k|$ factor normalises each mode's drive by its own residual, so a mode keeps receiving an $O(1)$ effective update until its residual is exhausted: a residual-gated energy injection in the sense of (A)/(C) that *sustains* tail modes rather than letting them decay at the gradient-flow rate. Adam thus combines timescale inversion ($\delta = 1$) with residual-gated sustenance—both of which raise the transient r relative to SGD.

Conjecture XIII.10 (Adam's edge is partly transient r-elevation)**.** *On targets with exploitable structure,* ADAMW *realises $r_{\mathrm{Adam}}(t) > r_{\mathrm{SGD}}(t)$ through the bulk of training, yielding a better finite-horizon scaling prefactor; asymptotically, absent $d^*_{\mathrm{loc}} < d^*$, both relax to $r = \frac{1}{2}$ (theorem V.2), consistent with theorem VII.1. The prediction is operational via theorem XIII.5: under matched Adam-vs-SGD runs one should observe a sustained gap in $\hat{r}(t)$ and a larger, slower-decaying $\left\|\dot{K}\right\|_F$ for Adam. A confirmed gap would make "Adam injects spectral energy" precise and would motivate an explicit, controllable injection (mechanism A) as a more direct and auditable substitute.*

F. Status of section XIII

Theorems: equilibrium relaxation (theorem XIII.1), the driven-NESS spectrum and its stability ceiling (theorems XIII.6 and XIII.7), and the no-shift property of positive preconditioning (theorem XIII.9). Heuristics: the Langevin reduction's white diffusion (31), the infinite-mode Onsager action (33), the anisotropic-noise tilt (section XIII-D3), and the log-only σ-surrogate (theorem XIII.5). Hypothesis: the Adam mechanism (37) and the r-gap (theorem XIII.10). All of these sit downstream of the *same* DMFT closure flagged in section XI-C: a rigorous closure of (18) would upgrade the heuristics to theorems. Crucially, nothing here assumes the spectral stationarity it characterises, and nothing weakens R1—which theorem XIII.8 localises to the relaxed worst-case asymptote rather than repeals. The constructive content is a reading of the two exits as one statement: acceleration is either a cheaper exponent bought by structure ($d^*_{\mathrm{loc}} < d^*$) or a prefactor bought by sustained work ($\sigma > 0$), and only the former survives the $D \to \infty$ limit on the worst-case ball.

XIV. Beyond the Barrier III: Cross-Entropy Loss and Discrete Data

The contact with language in section XII-D was made at the explicit grade of *heuristic*, and I said why twice: squared loss is not cross-entropy, and discrete tokens are not a manifold. These are gap (b) of section XI-F. This section closes both—not by assertion, but by isolating exactly what each gap costs and showing that, on a precisely named non-degeneracy domain, the parent exponents transport unchanged, while the genuinely new physics lives at a boundary that the worst-case barrier already forbids. The payoff is operational: once the data geometry is discrete, the intrinsic dimension d^* stops being a free parameter and becomes a *measurable* spectral observable, which lets one predict the compute-optimal split $(N^\star, D^\star)$ before the expensive run. I keep the discipline of section XI-A throughout: every claim is graded, and the two load-bearing assumptions are named, not smuggled.

A. Cross-entropy is a Fisher-weighted squared loss

Let the network emit a logit field $f(x) \in \mathbb{R}^m$ over $m = |\mathsf{V}|$ classes, with predictive law $p(x) = \mathrm{softmax}(f(x))$, and let the Bayes-optimal logit be $f^*(x)$, defined up to an additive constant per coordinate-block (the softmax gauge $f \mapsto f + c(x)\mathbb{1}$). The population cross-entropy decomposes exactly as

$$\mathcal{L}_{\mathrm{CE}}(f) = \underbrace{\mathbb{E}_x \, \mathrm{KL}\big(p^*(x) \,\|\, p(x)\big)}_{\text{excess loss}} + \underbrace{\mathbb{E}_x \, \mathbb{H}\big(p^*(x)\big)}_{= \mathbb{H}(y|x) =: L_0}, \tag{38}$$

so the *irreducible* term is the Bayes conditional entropy $L_0 = \mathbb{H}(y \mid x)$—the cross-entropy analogue of the floor in (1), absent from squared loss and here identified with a concrete data quantity.

Lemma XIV.1 (Fisher reduction of the excess loss)**.** *Write the logit error $e = f - f^*$. Then*

$$\mathrm{KL}\big(p^*(x) \,\|\, p(x)\big) = \tfrac{1}{2}\, e(x)^\top \mathsf{F}(x)\, e(x) + O\big(\|e(x)\|^3\big), \qquad \mathsf{F}(x) = \mathrm{diag}\big(p^*(x)\big) - p^*(x)\, p^*(x)^\top, \tag{39}$$

where $\mathsf{F}(x)$ is the multinomial Fisher information (equivalently the softmax Jacobian, equivalently the model's own predictive covariance). Consequently, to leading order around the Bayes logit, cross-entropy training is squared-error training in the Fisher metric $\langle e,\, e\rangle_{\mathsf{F}} := \mathbb{E}_x\, e^\top \mathsf{F} e$.

The capacity term of (8) is untouched by this change: it is the rich-regime implicit bias of the architecture (theorem IV.2), a property of the parametrisation $\lambda_k = w_k^2$ and the depth count $\nu = 1/L$, and is blind to which discrepancy functional sits on top of the features. Only the *reconstruction* term is loss-dependent, and theorem XIV.1 replaces the L^2 inner product by $\langle \cdot, \cdot \rangle_{\mathsf{F}}$. Diagonalising the feature operator in the F-orthonormal basis $\{\psi_k^{\mathsf{F}}\}$ and writing $\bar{a}_k^{\mathsf{F}} = \big\langle f^*,\, \psi_k^{\mathsf{F}} \big\rangle_{\mathsf{F}}$, theorem IV.1 becomes the Fisher-weighted cost $\sum_k (\bar{a}_k^{\mathsf{F}})^2/\lambda_k$, and the derived potential keeps its form,

$$V_{\mathrm{CE}}(\boldsymbol{\lambda}) = \sum_{k \geq 1} \frac{(\bar{a}_k^{\mathsf{F}})^2}{\lambda_k} + \mu \sum_{k \geq 1} \lambda_k^{\nu}, \tag{40}$$

identical to (8) under the single substitution $(\bar{a}_k, \psi_k) \mapsto (\bar{a}_k^{\mathsf{F}}, \psi_k^{\mathsf{F}})$.

B. The attractor and self-organised criticality survive (R2, R3)

Because (40) has the form of (8), the KKT stationarity $\partial V_{\mathrm{CE}}/\partial \lambda_k = 0$ reproduces the attractor of (10) verbatim,

$$\lambda_k^* \propto \big((\bar{a}_k^{\mathsf{F}})^2\big)^{1/(\nu+1)} \asymp k^{-b}, \qquad b = \frac{1 + 2t_{\mathsf{F}}}{\nu + 1}, \tag{41}$$

where t_{F} is the source exponent of f^* *in the Fisher metric*, i.e. $(\bar{a}_k^{\mathsf{F}})^2 \asymp k^{-(1+2t_{\mathsf{F}})}$. Whether self-organised criticality is preserved therefore reduces to a single question: *does the Fisher reweighting move the source exponent*, $t_{\mathsf{F}} \stackrel{?}{=} t$?

Proposition XIV.2 (Non-degeneracy $\Rightarrow$ exponents preserved)**.** *Suppose the target conditionals are uniformly non-degenerate: there is $p_0 > 0$ with $\min_a p_a^*(x) \geq p_0$ for a.e. x. Then on the gauge-orthogonal subspace $\{e : \mathbb{1}^\top e = 0\}$ the Fisher metric is uniformly equivalent to L^2,*

$$c_0(p_0)\ \|e\|_{L^2}^2 \ \leq\ \langle e,\, e\rangle_{\mathsf{F}} \ \leq\ \|e\|_{L^2}^2\,, \qquad c_0(p_0) \gtrsim p_0, \tag{42}$$

so the Fisher and L^2 Sobolev widths, and hence the source exponents, coincide in rate: $t_{\mathsf{F}} = t$ and $b = (1+2t)/(\nu+1)$ as in (10)*. Consequently theorems V.1 and V.2 hold for cross-entropy unchanged: $r(\nu) = t(\nu+1)/(1+2t)$, strictly increasing, with $r = \frac{1}{2}$ exactly at $\nu = 1/(2t)$, and over-alignment forbidden by the barrier (R1-CE below). Stationary feature learning under cross-entropy self-tunes to the same critical exponent.*

Sketch. $\mathsf{F}(x)$ is symmetric PSD with a single null direction $\mathbb{1}$ (the gauge); on its complement its eigenvalues lie in $[\,p_0(1-p_0),\, \frac{1}{4}\,]$ by the standard bound on the multinomial covariance, giving (42). Norm equivalence with x-independent constants preserves Kolmogorov and best-N-term widths up to constants [16], hence preserves the power-law exponents; only the prefactor μ in (40) absorbs the $O(1)$ distortion, exactly as the nonlinearity did in theorem XI.3. ☐

Remark XIV.3 (The deterministic limit: where cross-entropy genuinely deviates)**.** *The one place the squared-loss picture and cross-entropy part company is the* near-deterministic *regime $p^*_{\min}(x) \to 0$ on a set of positive measure (highly predictable contexts, the low-entropy tail of language). There $c_0(p_0) \to 0$,* (42) *degenerates, the Fisher metric collapses along the saturating coordinates, and t_{F} acquires a boundary-layer correction $t_{\mathsf{F}} = t + \Delta(p_0)$ with $\Delta \to 0$ as $p_0 \to 0$ only through a logarithmic margin, reflecting the* KL*/softmax blow-up near the simplex boundary. By the same localisation as theorem XIII.8, this moves a* prefactor *and the pre-asymptotic exponent realised at finite D on easier-than-worst-case targets; it cannot move the worst-case asymptotic exponent, which R1-CE pins. Status:* heuristic*—the boundary-layer expansion is not closed here, and it inherits the same DMFT-closure obstruction as section XI-C.*

C. The barrier transports (R1)

Theorem XIV.4 (Cross-entropy minimax barrier)**.** *Let the conditional law of $y \mid x$ have logit f^* ranging over $\{\|f^*\|_{\mathcal{H}^s(\mathcal{M})} \leq R\}$ with densities bounded in $[p_0, 1-p_0]$, $p_0 > 0$, on a manifold of dimension d^*. For any estimator $\hat{p}_D$ from D i.i.d. samples,*

$$\inf_{\hat{p}_D}\ \sup_{\|f^*\|_{\mathcal{H}^s} \leq R}\ \mathbb{E}\ \mathrm{KL}\big(p^* \,\|\, \hat{p}_D\big)\ \gtrsim\ D^{-\beta_0}, \qquad \beta_0 = \frac{2s}{2s+d^*}, \tag{43}$$

the same *exponent as* (12)*. Excess cross-entropy obeys the same bound, since by* (38) *it equals $\mathbb{E}_x\,\mathrm{KL}$.*

Sketch. On the bounded-density class the three risks are uniformly comparable, $\mathrm{KL} \asymp \mathrm{H}^2 \asymp \|\cdot\|_{L^2}^2$ [5], with constants depending only on p_0. The Fano/hypercube construction of theorem VI.1 then applies verbatim: the per-sample KL separation of the $M \asymp \delta^{-d^*/s}$ perturbations is $\asymp \varepsilon^2$, and optimising returns $\varepsilon^2 \asymp D^{-2s/(2s+d^*)}$. The bounded-density hypothesis is the exact density-estimation counterpart of the non-degeneracy of theorem XIV.2; drop it and the comparison constants diverge—the same boundary effect as theorem XIV.3. ☐

R1–R4 therefore hold for cross-entropy on the non-degenerate class, with the identical exponent algebra of section XI-A. The empirical floor L_0 in (1) is no longer a fitting constant but the Bayes conditional entropy $\mathbb{H}(y \mid x)$, and the regime map of section IX stands unchanged with L_0 so interpreted.

D. Discrete data: the token graph and its Weyl law

A corpus is not an $L^2(\mathcal{M})$ field on a smooth manifold but a distribution over discrete sequences. I replace $\mathcal{M}$ by a weighted *token graph* $\mathcal{G} = (\mathsf{V}, \mathsf{E})$: vertices V are tokens (or fixed-length context states), edge weights W_{ij} are (symmetrised) co-occurrence affinities. Let $\Delta_{\mathcal{G}} = I - D^{-1/2} W D^{-1/2}$ be the normalised graph Laplacian, with spectrum $0 = \mu_0 \le \mu_1 \le \cdots \le \mu_{|\mathsf{V}|-1} \le 2$ and eigenvectors $\{\psi_k\}$. Define the *discrete Sobolev* space by functional calculus,

$$\|f\|^2_{\mathcal{H}^s(\mathcal{G})} := \sum_k (1+\mu_k)^s \, \langle f,\, \psi_k\rangle^2 \,, \tag{44}$$

the exact discrete analogue of $\sum_k (1+\mu_k)^s \bar{a}_k^2$ from section II. The Weyl law of (2), however, is a continuum theorem and does not hold for an arbitrary graph; it must be *earned*. The condition under which it survives is the discrete form of the manifold hypothesis.

Assumption XIV.5 (Graph-Laplacian consistency; discrete manifold hypothesis)**.** *The co-occurrence graph is a consistent discretisation of a latent metric-measure space of intrinsic dimension d^*: as $|\mathsf{V}| \to \infty$ with a suitably scaled affinity bandwidth, $\Delta_{\mathcal{G}}$ converges spectrally to the Laplace–Beltrami operator of that space [27], [28]. Equivalently, the empirical graph spectrum obeys a Weyl law on a scaling window,*

$$\mu_k^{\mathcal{G}} \asymp k^{2/d^*}, \qquad 1 \ll k \ll |\mathsf{V}| \,, \tag{45}$$

and the Sobolev kernel eigenvalues decay as $\sigma_k \asymp k^{-b_0}$, $b_0 = 2s/d^*$, on that window.

Theorem XIV.6 (Discrete R1–R4, conditional on consistency)**.** *Under theorem XIV.5, the entire parent chain transports with d^* the latent graph dimension and $\{\psi_k\}$ the graph-Laplacian eigenbasis: the minimax barrier* (12) *holds for $\mathcal{H}^s(\mathcal{G})$ targets (R1), the approximation rate is $\alpha = 2s/d^*$ (R4), the attractor* (10) *and criticality $r = \frac{1}{2}$ hold (R2, R3), and the crossover* (16) *is unchanged. The finite vocabulary supplies a spectral (UV) cutoff at $k_{\max} \asymp |\mathsf{V}|$:* (45) *is valid only below it, and structure finer than the graph's resolution is unresolvable. This cutoff is a second, geometry-side contribution to the irreducible floor of* (38)*, additive with $L_0 = \mathbb{H}(y \mid x)$.*

Sketch. Theorem XIV.5 furnishes the only continuum input the parent proofs use—the Weyl exponent in (2) and the eigenbasis. With (45) every spectral sum in sections V, VII and VIII is the same integral over the same scaling window, truncated at $k_{\max}$; the truncation changes the regime boundary but not the exponents, and contributes a $k_{\max}$-dependent constant to the floor. The barrier theorem VI.1 uses only the counting function $\#\{k : \mu_k \le \Lambda\} \asymp \Lambda^{d^*/2}$, which (45) reproduces below $k_{\max}$. □

Remark XIV.7 (d^* becomes measurable)**.** *In the continuum theory d^* was a property* assumed *of $\mathcal{M}$ (section II), and t was a free parameter ((3)). On the token graph, d^* is the slope of* (45)*, computable directly from the corpus co-occurrence statistics, and $b_0 = 2s/d^*$ is the slope of the trained kernel. Discreteness thus makes d^* computable directly from the corpus rather than postulated, and—because the two slopes are measured independently—falsifiable (section XIV-E).*

E. Measuring d^* and d^*_{loc}, and predicting $(N^\star, D^\star)$ a priori

Theorem XIV.7 makes the effective-dimension machinery of section XII operational. I give the protocol and grade its output.

Proposition XIV.8 (Effective-dimension measurement protocol)**.** *Fix a target corpus. The following recovers $(d^*, s, d^*_{\mathrm{loc}})$ and hence the compute-optimal allocation, at a cost dominated by one small* probe *model.*

(M1) ***Global** d^* **(data side).** Build the symmetrised co-occurrence graph $\mathcal{G}$, form $\Delta_{\mathcal{G}}$, and regress $\log \mu_k$ on $\log k$ over the scaling window of* (45)*; the slope is $2/d^*$.*

(M2) *b_0 **and** s **(model side).** Train a small probe; form its post-training empirical kernel as in theorem X.1; read $\hat{\lambda}_k \asymp k^{-b_0}$. With d^* from (M1), $s = \frac{1}{2} b_0 d^*$.*

(M3) ***Local** d^*_{loc} **(compositional side).** Apply an intrinsic-dimension estimator—TwoNN [29] or the MLE of [30]—to the probe's* intermediate-layer *activations, level by level. By theorem XII.6, $d^*_{\mathrm{loc}} = \max_i d_i$ is the largest per-level estimate; the per-layer profile localises the binding constituent.*

Proposition XIV.9 (A priori compute-optimal split)**.** *Given (d^*, s) from theorem XIV.8, the parent regime analysis (section IX) predicts,* before *the full run, the knee $D_{\mathrm{cross}} = N^{(2s+d^*)/d^*}$ and the frontier allocation*

$$N^\star \propto C^{\,d^*/[2(s+d^*)]}, \qquad D^\star \propto C^{\,(2s+d^*)/[2(s+d^*)]}, \tag{46}$$

with the data-heavy tilt set by $\alpha - \beta_0 = 4s^2/[d^*(2s+d^*)]$ (theorem VIII.3); replacing d^* by the measured d^*_{loc} from (M3) gives the structure-corrected matched depth $L^\star = 2s/d^*_{\mathrm{loc}}$ (theorem XII.12). Status: theorem conditional on the measurement*—the formulae are R1/R4 algebra; their predictive validity for a given corpus stands or falls with theorem XIV.5.*

Remark XIV.10 (A falsifiable consistency check)**.** *Steps (M1) and (M2) measure d^** twice*: once from the static graph spectrum, once (via s, anchored independently, e.g. by smoothness of the embedding map) from the trained-kernel slope $b_0 = 2s/d^*$. Under theorem XIV.5 the two agree. Disagreement is not noise to be averaged away—it falsifies the discrete manifold hypothesis for that corpus, exactly as the two-d^*_{loc} consistency test of section XII-G certifies or refutes the effective-dimension picture. This upgrades the language contact of section XII-D from heuristic analogy to a measurement with a refutation condition.*

F. Status of section XIV

In the grading discipline of section XI-A:

a) Theorems (under stated hypotheses).: The Fisher reduction theorem XIV.1 (a second-order expansion, exact to $O(e^3)$); the norm-equivalence theorem XIV.2 and the consequent transport of R2/R3 to cross-entropy on the non-degenerate class; the cross-entropy minimax barrier theorem XIV.4 (R1-CE), by KL/Hellinger/L^2 comparability on bounded densities [5]; and theorem XIV.6 (discrete R1–R4) *conditional on theorem XIV.5*. Each reduces to its parent statement when the Fisher metric is the identity and the graph is the manifold, and the exponent identities are unchanged.

b) Heuristics.: The deterministic-limit correction $t_{\mathsf{F}} = t + \Delta(p_0)$ (theorem XIV.3); and the identification of the two irreducible contributions—$L_0 = \mathbb{H}(y \mid x)$ from cross-entropy and the $|\mathsf{V}|$-cutoff from discreteness—as the two sources of the floor in (1), stated at the level of mechanism rather than a closed rate.

c) Hypotheses (named, open).: Two, and only two, carry the section. **(H-CE)** uniform non-degeneracy of the target conditionals, $p^*_{\min} \geq p_0 > 0$; it is what makes the Fisher metric norm-equivalent to L^2 and is violated precisely in the near-deterministic tail. **(H-graph)** graph-Laplacian spectral consistency (theorem XIV.5), the discrete manifold hypothesis, under which the Weyl law and hence d^* are well-defined. Neither bears on R1–R4 of the parent paper; both sit strictly downstream, and—unlike the heuristic of section XII-D—both are *checkable*: (H-CE) by the empirical next-token entropy floor, (H-graph) by the straightness of (45) on a log–log plot.

d) Decisive tests.: Three, extending theorem X.1 and section XII-G. *(i) Entropy floor:* the fitted L_0 in (1) should match the measured conditional entropy $\mathbb{H}(y \mid x)$. *(ii) Weyl straightness:* the graph spectrum (45) should be a clean power law over the scaling window, with slope $2/d^*$. *(iii) Double-d^* consistency* (theorem XIV.10): the data-side and model-side d^* must agree, on pain of falsifying (H-graph). What was qualitative contact with language in section XII-D is, on this section's domain, a quantitative and refutable program: the contact is now at the rate of theorem wherever (H-CE) and (H-graph) hold, and exactly localised where they fail.

XV. Beyond the Barrier IV: The DMFT–Adam Closure Theorem

The dependency chain (19) names three open links: the non-perturbative closure of Γ_k (section XI-C), the eigenbasis-stability bound theorem XI.1 consumed by the conditional nonlinear H1 (theorem XI.2), and the status of Adam, carried only as a conjecture (theorem XIII.10). This section proves a theorem that closes them in the regime where they are decidable, and is explicit about the one place where they are not.

The organising observation is a separation the earlier sections left implicit. *The Adam window lives in the weak-coupling phase.* Alignment (theorem XI.1) is, by its own statement, the requirement that the inter-mode coupling be summable, $\sup_\tau \sum_k |\Gamma_k| < \infty$; that is weak coupling. The elevation $r > \frac{1}{2}$ that Adam produces does *not* come from the memory kernel—it comes from the *external*, non-conservative injection (37) sitting on top of an aligned, weakly coupled flow. The memory-kernel-driven overshoot of section XI-C's strong-coupling branch is a *separate* mechanism, and it is the one that is genuinely open. I therefore prove everything the Adam result needs inside weak coupling, where the Gaussianity supplied by theorem III.1 makes the closure rigorous, and I isolate the strong-coupling closure as the single irreducible residue.

Within this section τ denotes training time and t retains its meaning as the source exponent of f^* ((3)), worst case $t = s/d^*$; the limit is $n \to \infty$ at fixed depth L, so theorem III.1 furnishes Gaussian preactivations and the spectral flow (6).

Definition XV.1 (Weak-coupling phase)**.** *Let $\mathcal{J}$ be the bare inter-mode coupling operator of* (6) *(the off-diagonal Hessian of the flow* (5)*) and $\kappa_k = 2\bar{a}_k^2/(\lambda_k^*)^3$ the diagonal restoring curvature at the attractor* (10)*. The flow is in the* weak-coupling phase *on a time interval if*

$$\|\mathcal{J}\|_{\mathrm{op}}^2 \;<\; \eta \left(\inf_k \kappa_k\right)^2 \qquad (\textit{sub-criticality}), \tag{47}$$

a condition on bare*, computable quantities (no reference to the closure's output). Its complement is the strong-coupling phase.*

A. Part 1: rigorous DMFT closure in weak coupling

The closure has two ingredients: a concentration result that makes the self-consistent order parameters deterministic, and a fixed-point result that makes them exist and preserves the phase structure. Both are proved; the first rests on the Gaussianity of theorem III.1, not on a new hypothesis.

a) Spectral shells.: Partition the spectrum into dyadic shells $\mathcal{S}_j = \{k : 2^j \le k < 2^{j+1}\}$, $|\mathcal{S}_j| \asymp 2^j \to \infty$. Within a shell the envelope $\lambda_k \asymp k^{-b}$ varies by a bounded factor 2^b, so modes are exchangeable *up to a known deterministic profile*; this is the heterogeneous (labelled) mean-field setting, not full exchangeability.

Lemma XV.2 (Diagonal-projection decorrelation; exact under Gaussianity)**.** *Let $z_k = \langle \psi_k,\, \phi(\mathbf{h}^{(L)}) \rangle$ be the projection of the feature field onto ψ_k. By theorem III.1 the field is Gaussian with $\mathbb{E}[z_j z_k] = \sigma_k \delta_{jk}$ in the eigenbasis. Let $u_k = \psi_k^\top(\hat{\Sigma}_D - \Sigma)\psi_k = \frac{1}{D}\sum_{i \le D}(z_{k,i}^2 - \mathbb{E} z_k^2)$ be the empirical fluctuation that drives mode k in* (31)*. Then, by Wick's theorem,*

$$\mathrm{Cov}(u_j, u_k) = \frac{1}{D}\,\mathrm{Cov}(z_j^2, z_k^2) = \frac{2}{D}\,\mathbb{E}[z_j z_k]^2 = \frac{2}{D}\,\sigma_k^2\,\delta_{jk}. \tag{48}$$

The per-mode driving fluctuations are therefore exactly uncorrelated across distinct modes *(and, being jointly Gaussian, independent to leading order), with variance $O(1/D)$.*

Proof. z is centred Gaussian (subtract the deterministic mean field), so $\mathrm{Cov}(z_j^2, z_k^2) = 2\,\mathbb{E}[z_j z_k]^2$ by Isserlis/Wick; orthonormality of $\{\psi_k\}$ diagonalises Σ, giving $\mathbb{E}[z_j z_k] = \sigma_k \delta_{jk}$. Averaging D i.i.d. samples divides the covariance by D. □

Lemma XV.3 (Within-shell concentration / conditional propagation of chaos)**.** *In the weak-coupling phase* (47)*, the linearised fluctuations $\delta\lambda_k = \lambda_k - \lambda_k^*$ obey a stable Ornstein–Uhlenbeck system with rate $\eta\kappa_k$ and the independent forcing of theorem XV.2 perturbed by the sub-critical coupling $\mathcal{J}$. The empirical shell order parameters*

$$\mathsf{C}_j(\tau, \tau') = \frac{1}{|\mathcal{S}_j|}\sum_{k \in \mathcal{S}_j} \delta\lambda_k(\tau)\delta\lambda_k(\tau'), \qquad \mathsf{R}_j(\tau, \tau') = \frac{1}{|\mathcal{S}_j|}\sum_{k \in \mathcal{S}_j} \frac{\delta\,\delta\lambda_k(\tau)}{\delta\xi_k(\tau')}, \tag{49}$$

concentrate on their means with variance

$$\mathrm{Var}(\mathsf{C}_j),\ \mathrm{Var}(\mathsf{R}_j)\ =\ O\Big(\frac{1}{|\mathcal{S}_j|}\cdot\frac{1}{1-\|\mathcal{J}\|_{\mathrm{op}}^2/(\eta\inf_k\kappa_k^2)}\Big)\ \xrightarrow[|\mathcal{S}_j|\to\infty]{}\ 0. \tag{50}$$

Proof sketch. In weak coupling the linearised single-mode equation is the stable OU process $\dot{\delta\lambda}_k = -\eta\kappa_k\,\delta\lambda_k + (\mathcal{J}\,\delta\lambda)_k + \xi_k + \sqrt{2D_k}\dot{W}_k$, whose stationary response is bounded by $1/(\eta\inf_k\kappa_k)$; sub-criticality (47) makes the Neumann series in $\mathcal{J}$ converge, so cross-mode covariances decay geometrically in coupling order. Within a shell the $|\mathcal{S}_j|$ summands of (49) are then independent up to the geometric coupling correction (theorem XV.2 gives independence at zeroth order), and the variance bound (50) is the law of large numbers for weakly dependent bounded variables (e.g. the classical mean-field propagation-of-chaos estimate of Sznitman in the labelled/heterogeneous form). The deterministic envelope λ_k^* needs no concentration: it is the $n\to\infty$ limit of theorem III.1. □

Proposition XV.4 (Self-consistent closure, explicit in weak coupling)**.** *Under* (47) *the effective single-mode dynamics is the OU process of theorem XV.3 with coloured noise of self-consistent covariance, closed by*

$$\mathsf{C}_j(\tau,\tau') = \langle\delta\lambda(\tau)\delta\lambda(\tau')\rangle,\quad \mathsf{M}_j = \big[\mathcal{J}\,\mathsf{R}\,\mathcal{J}^\top\big]_j,\quad \mathsf{R}_j(\tau,\tau') = \Big\langle\tfrac{\delta\,\delta\lambda(\tau)}{\delta\xi(\tau')}\Big\rangle. \tag{51}$$

This is (18) *with $M=\mathsf{M}_j$, $C=\mathsf{C}_j$* determined*, not whitened; because the process is Gaussian/OU the system* (51) *is linear and solvable in closed form by Fourier transform in $\tau-\tau'$.*

Proposition XV.5 (Existence, uniqueness, and phase preservation)**.** *The map $\mathcal{F}:(\mathsf{M},\mathsf{C})\mapsto(\mathsf{M},\mathsf{C})$ of* (51) *is a contraction in the norm $\|\mathsf{M}\|_{\mathrm{op}}/\inf_k\kappa_k$ on the weak-coupling phase* (47)*, with a unique fixed point continuously connected to the white-noise solution; it loses contractivity exactly on the boundary of* (47)*. Hence the closure has the* same *two-branch topology as the perturbative classification of section XI-C: weak coupling $\Rightarrow r\to\frac{1}{2}$ (recovering theorem V.2); the boundary of* (47) *is the only bifurcation, so colouring the noise displaces the critical coupling by an $O(1)$ factor but creates no intermediate phase.*

Proof. The response of a stable OU process is bounded by the inverse of its rate, so $\|\mathsf{R}\|\le 1/(\eta\inf_k\kappa_k)$; then $\|\mathsf{M}\| = \big\|\mathcal{J}\mathsf{R}\mathcal{J}^\top\big\| \le \|\mathcal{J}\|^2/(\eta\inf_k\kappa_k)$, which is $<\inf_k\kappa_k$ precisely under (47). On that set $\mathcal{F}$ is linear in a strict contraction, so Banach's theorem gives existence and uniqueness; contractivity fails exactly when $\|\mathsf{M}\| = \inf_k\kappa_k$, the sign change of the activating-mode OU rate. No third branch can appear because a linear contraction has a single fixed point. □

Remark XV.6 (The single open residue: strong coupling)**.** *Outside* (47) *the single-mode process is no longer OU—large excursions $\lambda_k\gg\lambda_k^*$ probe the nonlinear potential* (8)*, the fluctuations are non-Gaussian, and theorem XV.3 does not apply. A rigorous closure there is the propagation-of-chaos problem for an* evolving, strongly self-coupled *spectral field, which is open as it is for DMFT of feedback-driven kernels in general (the Ben Arous–Dembo–Guionnet programme is rigorous for fixed quenched disorder, not for a kernel the dynamics regenerates). This residue carries the memory-kernel-driven overshoot of section XI-C;* it is not used below*, because the Adam window is a weak-coupling phenomenon. I mark it open and do not stretch the weak-coupling proof to cover it. Section XV-D returns to this residue and closes it on the entire alignment window $[0,T^\star]$ (and, under a decaying schedule, for all time); only a post-window glassy tail, asymptotically inert by R1, remains genuinely open.*

B. Part 2: the Adam alignment window, derived

I now *derive* theorem XI.1 on an explicit interval, correcting its statement on the way: the true alignment functional carries spectral-gap denominators, which the heuristic equivalence in theorem XI.1 omits and which are what make the window finite.

Lemma XV.7 (Gap-gated rotation; correction to theorem XI.1)**.** *With eigenpairs $(\lambda_k(\tau), \hat{\psi}_k(\tau))$ of $K^{(L)}(\tau)$, first-order perturbation theory gives the rotation generator and alignment functional*

$$A_{jk} = \frac{\left\langle \hat{\psi}_j,\, (\partial_\tau K^{(L)})\hat{\psi}_k \right\rangle}{\lambda_k - \lambda_j} \;(j \neq k), \qquad \Theta(\tau) := \sum_k \left\| \partial_\tau \hat{\psi}_k \right\|^2 = \sum_{j \neq k} A_{jk}^2. \tag{52}$$

The numerator is the off-diagonal coupling, of scale $|\Gamma_k| = O(k^{-\gamma})$; the denominator is the gap. The correct alignment functional is Θ, not $\sum_k |\Gamma_k|$: *since adjacent gaps shrink as $\lambda_k - \lambda_{k+1} \asymp b\, k^{-b-1}$, Θ can diverge even when $\sum_k |\Gamma_k| < \infty$. theorem XI.1 should be read as $\sup_\tau \Theta(\tau) < \infty$.*

Lemma XV.8 (Explicit misalignment growth and the window $T^\star$)**.** *In the weak-coupling phase, with $\lambda_k \asymp k^{-b}$, $b = (1+2t)/(\nu+1)$, $\nu = 1/L$, and the gradient-flow resolution front $k_{\max}(\tau) \asymp (\int_0^\tau \eta)^{1/b_0}$, $b_0 = 2s/d^*$, the adjacent tail pairs dominate* (52) *and*

$$\Theta(\tau) \;\asymp\; \frac{1}{b^2(2b+3-2\gamma)}\; k_{\max}(\tau)^{\,2b+3-2\gamma} \;\asymp\; \frac{1}{b^2(2b+3-2\gamma)} \Big(\int_0^\tau \eta \Big)^{(2b+3-2\gamma)/b_0}. \tag{53}$$

Let $\Theta_\dagger$ be the ceiling at which the diagonal-in-eigenbasis approximation of section XIII-E (equivalently the descent bound $\eta_k \kappa_k \leq 2$ of theorem XIII.7) first fails. Then $\Theta(\tau) \leq \Theta_\dagger$, i.e. theorem XI.1 holds, on $[0, T^\star]$ with

$$\boxed{\int_0^{T^\star} \eta(s)\, ds \;\asymp\; \big[\, b^2(2b+3-2\gamma)\, \Theta_\dagger \,\big]^{\,b_0/(2b+3-2\gamma)}, \qquad b_0 = \tfrac{2s}{d^*}, \quad b = \tfrac{(1+2t)L}{L+1}.} \tag{54}$$

Proof. For $j = k \pm 1$, $A_{k,k+1}^2 \lesssim |\Gamma_k|^2 / (\lambda_k - \lambda_{k+1})^2 \asymp k^{-2\gamma}/(b^2 k^{-2b-2}) = b^{-2} k^{2b+2-2\gamma}$; non-adjacent pairs have larger gaps and contribute subdominantly. Summing $\sum_{k \leq k_{\max}} k^{2b+2-2\gamma} \asymp k_{\max}^{2b+3-2\gamma}/(2b+3-2\gamma)$ (the exponent exceeds -1 for any $\gamma < b + \frac{3}{2}$) gives (53). Substituting the front $k_{\max}(\tau) \asymp (\int_0^\tau \eta)^{1/b_0}$ (each mode relaxes at NTK rate $\eta\sigma_k \asymp \eta k^{-b_0}$, so k is resolved when $\int_0^\tau \eta \gtrsim k^{b_0}$) and solving $\Theta(T^\star) = \Theta_\dagger$ yields (54). □

Remark XV.9 (Dependence on (d^*, s, L) and the schedule; finite vs. infinite window)**.** *By* (54) *the alignment budget $\int_0^{T^\star} \eta$ is decreasing in b and the exponent $b_0/(2b+3-2\gamma)$ controls the trade with the schedule. Through $b = (1+2t)L/(L+1)$ and $t = s/d^*$, smoother targets (larger s/d^*) and shallower nets (smaller L) give larger b hence a* shorter *window: alignment is lost sooner exactly where the spectrum is steepest and tail gaps smallest. For a constant rate the window is finite, $\eta_0 T^\star \asymp [b^2(2b+3-2\gamma)\Theta_\dagger]^{b_0/(2b+3-2\gamma)}$. For a decaying schedule $\eta(\tau) = \eta_0(1+\tau/\tau_0)^{-p}$ the budget $\int_0^\infty \eta$ is finite when $p > 1$; if that finite budget is below the right-hand side of* (54) *then $T^\star = \infty$ and alignment is sustained for all time. This is the precise sense in which the learning-rate schedule sets the window.*

Theorem XV.10 (Adam realises $r > \frac{1}{2}$ on the window)**.** *Work in the weak-coupling phase* (47) *and assume $t > \frac{1}{2}$. Let $\tau_{\rm relax} \asymp 1/(\eta\, \kappa_{\rm head})$ be the head relaxation time. If the window is long enough, $T^\star > \tau_{\rm relax}$ (a checkable inequality via* (54)*), then on $[\tau_{\rm relax}, T^\star]$ the diagonal approximation of section XIII-E holds (theorem XV.8), the Adam step* (37) *acts as a residual-gated tail injection of effective exponent $\gamma_{\rm eff} \in (0,\, 1 - \frac{1}{2t})$ within the stability ceiling $\gamma < 1 - \nu$ of theorem XIII.7, and theorem XIII.6 applies modewise to give*

$$r_{\rm Adam}(\tau) = \frac{t\,(2 - \gamma_{\rm eff})}{1+2t} \;>\; \tfrac{1}{2}. \tag{55}$$

For $t \leq \frac{1}{2}$, $r_{\rm Adam} \uparrow \frac{1}{2}$ without crossing it. For $\tau > T^\star$ the eigenbasis rotates ($\Theta > \Theta_\dagger$), the modewise product structure of theorem XI.2 fails, and the relaxed asymptote is $r = \frac{1}{2}$ (theorem V.2,theorem XIII.8) unless the target supplies $d^*_{\rm loc} < d^*$.

Proof. theorem XV.8 certifies $\Theta \leq \Theta_\dagger$ on $[0, T^\star]$, the hypothesis under which (37) is exact and theorem XI.2 holds, so the spectrum stays co-diagonal with $\{\psi_k\}$ and the per-mode picture of theorem XIII.6

is valid. Adam's preconditioner is mechanism (B) with $\delta = 1$ plus a residual-gated non-conservative drive $F_k \propto \lambda_k^{-\gamma_{\text{eff}}}$ (section XIII-E); theorem XIII.6 with $\gamma \rightsquigarrow \gamma_{\text{eff}}$ gives (55). The driven NESS is reached after τ_{relax}, which is inside the window by hypothesis; $r_{\text{Adam}} > \frac{1}{2} \Leftrightarrow \gamma_{\text{eff}} < 1 - \frac{1}{2t}$ needs $t > \frac{1}{2}$ (theorem XIII.6); the ceiling $\gamma_{\text{eff}} < 1 - \nu$ is the descent bound defining $\Theta_\dagger$. Weak coupling is consistent with the drive because F is external and does not enter $\mathcal{J}$, so (47) and the NESS coexist. □

Remark XV.11 (Grading of the closure)**.** *theorem XV.8 derives theorem XI.1 on* $[0, T^\star]$ *from the gap-gated inequality* (53)*—a consequence of the Part 1 closure (which fixes the coupling scale* $|\Gamma_k| = O(k^{-\gamma})$ *through* $\mathsf{M} = \mathcal{J}\mathsf{R}\mathcal{J}^\top$*)—without ever assuming spectral stationarity. This forms the bridge DMFT* $\Rightarrow$ $theorem\ XI.1 \Rightarrow$ *nonlinear H1 of* (19)*, now quantitative rather than postulated.*

C. Part 3: tightness at fixed entropy production (unconditional)

This part uses neither weak coupling nor Gaussianity: it is a variational identity for any NESS. Let $\bar{\lambda}_k = \mathbb{E}_{P_{\text{ness}}}[\lambda_k] \asymp k^{-b}$, set the tail-flattening functional $\Phi = -b$ (so $r = t/b$ increases with Φ), and let $w_k = \delta\Phi/\delta\bar{\lambda}_k$ be its functional gradient; the steady current transports spectral mass at rate $\dot{\Phi} = \langle w,\, J\rangle$.

Theorem XV.12 (Tight no-free-lunch frontier)**.** *For every steady, divergence-free current J ($\sum_k \partial_{\lambda_k} J_k = 0$) of entropy-production rate $\sigma = \sum_k J_k^2/(D_k P_{\text{ness}})$ (theorem XIII.3),*

$$\dot{\Phi} = \langle w,\, J\rangle \;\le\; \sqrt{\sigma}\, \Big(\textstyle\sum_k D_k P_{\text{ness}}\, w_k^2\Big)^{1/2}, \qquad \text{equality iff } J_k \propto D_k P_{\text{ness}}\, w_k. \tag{56}$$

The equality-case current is monotone in the spectral tail (as w_k is), and the drift that generates it is, up to the $D_k P_{\text{ness}}$ dressing, the tail-pumping injection $F_k \propto \lambda_k^{-\gamma}$ of mechanism (A)*, section XIII-D1. Hence at fixed σ the maximal sustainable r equals r_{drive} of* (36)*, and $\beta(r_{\text{drive}})$ of* (13) *is the* tight *estimation ceiling: no first-order optimiser sustaining the same σ realises a larger r, hence none attains a smaller loss exponent. The map $\sigma \mapsto \beta(r_{\text{drive}}(\sigma))$ is the no-free-lunch frontier.*

Proof. eq. (56) is Cauchy–Schwarz in the Onsager/Otto inner product $\langle u,\, v\rangle_\sigma = \sum_k u_k v_k/(D_k P_{\text{ness}})$, in which $\sigma = \langle J,\, J\rangle_\sigma$ is the squared current norm (equivalently the squared $\mathcal{W}_2$-speed of the generating flow, by Benamou–Brenier). Writing $\langle w,\, J\rangle = \langle DP_{\text{ness}}\, w,\, J\rangle_\sigma \le \|DP_{\text{ness}}\, w\|_\sigma \sqrt{\sigma}$ gives the bound, with equality iff $J \parallel DP_{\text{ness}}\, w$, i.e. the current points along the metric gradient of Φ. Integrating that velocity field against the continuity equation identifies the generating drift with the monotone tail force of (A). Since $\dot{\Phi}$ caps the attainable b, hence $r = t/b$, and the cap is met by (A), the value is r_{drive} and $\beta(r_{\text{drive}})$ is tight by (13). □

Corollary XV.13 (Adam cannot beat explicit injection at equal cost)**.** *Whatever γ_{eff} Adam realises on its window (theorem XV.10), its sustained r at entropy-production rate σ cannot exceed $r_{\text{drive}}(\sigma)$ of an explicit injection at the same σ. Adam is at best an implicit realisation of mechanism (A); the explicit, auditable injection of section XIII-D1 dominates it cost-for-cost—the substitution proposed in theorem XIII.10.*

D. Closing the strong-coupling residue

Theorem XV.6 left one residue: the closure outside the weak-coupling phase (47). I now close it on the interval that carries every result of this section, and reduce the remainder to a single, asymptotically inessential tail with a named obstruction. The mechanism is that *alignment loss and sub-criticality loss are the same event*: both are driven by the growth of the off-diagonal coupling, so the window on which alignment holds is automatically a window on which weak coupling holds. There is therefore no gap between "aligned" and "weakly coupled" to fall through.

a) The spectral flow as a heterogeneous, non-autonomous McKean–Vlasov system.: Treat each mode as a particle carrying a deterministic *type* $\theta_k = k$ (its spectral position), with empirical type-and-state measure

$$\mu_\tau^N = \frac{1}{N}\sum_{k\le N}\delta_{(\theta_k,\lambda_k(\tau))}, \qquad \mu_\tau^N \xrightarrow{N\to\infty} \mu_\tau, \tag{57}$$

where the type density is the dyadic-shell profile of theorem XV.3. The drift of λ_k in (31) depends on the others only through μ_τ^N via the coupling Γ, and the dependence is *time-varying* because V_{tot} and the spectrum evolve. This amounts to a *heterogeneous* (typed) McKean–Vlasov diffusion with *non-autonomous* interaction. Let

$$\ell(\tau) := \mathrm{Lip}_{\mathcal{W}_2}\big(\mu\mapsto\Gamma[\mu](\tau)\big) \;=\; \frac{\|\mathsf{M}(\tau)\|_{\text{op}}}{\inf_k \kappa_k} \;\le\; \frac{\|\mathcal{J}(\tau)\|_{\text{op}}^2}{\eta\,(\inf_k \kappa_k)^2}, \qquad \tau_c := \inf\Big\{\tau : \textstyle\int_0^\tau \ell(s)\,ds = \infty\Big\}, \tag{58}$$

the interaction's Wasserstein-Lipschitz modulus and the first time its time-integral diverges. Sub-criticality (47) is exactly $\ell(\tau) < 1$.

Theorem XV.14 (Non-autonomous propagation of chaos up to τ_c; route 1)**.** *Assume the type density of* (57) *is regular (a bounded shell profile) and the per-mode coefficients are Lipschitz, as holds for the OU-linearised flow of theorem XV.3. Then for every $\tau < \tau_c$ the empirical measure concentrates,*

$$\mathcal{W}_2\big(\mu_\tau^N,\mu_\tau\big) \;=\; O\Big(N^{-1/2}\exp\!\textstyle\int_0^\tau \ell(s)\,ds\Big), \qquad \textit{uniformly on } [0,\tau], \tag{59}$$

and the self-consistent closure (51) *holds with no white-noise approximation. The sharp entropic rate $O(N^{-1})$ of the hierarchy method [31] applies under the same integrated-Lipschitz condition; the heterogeneity (continuum of types) is handled by the beyond-exchangeability framework of [32].*

Proof sketch. Bootstrap: validity of the OU-linearised flow on $[0,T^\star]$. The hypothesis of the theorem requires the per-mode coefficients to be Lipschitz, a condition the OU-linearised flow of theorem XV.3 satisfies—but only where the linearisation itself is valid, i.e. where $\delta\lambda_k = \lambda_k - \lambda_k^*$ is small relative to λ_k^*. I verify this by a bootstrap on $[0,T^\star]$.

(i) At $\tau = 0$, $\Theta(0) = 0$ because the kernel starts at its OU fixed point $\boldsymbol{\lambda}^*$ (the attractor (10) is the initialisation of the spectral flow in the mean-field limit $n\to\infty$), so all fluctuations $\delta\lambda_k(0) = 0$ and the linearisation is exact.

(ii) $\Theta(\tau)$ is monotone increasing along the resolution front (theorem XV.8), and $T^\star := \inf\{\tau : \Theta(\tau) = \Theta_\dagger\}$ by definition, so $\Theta(\tau) < \Theta_\dagger$ for all $\tau\in[0,T^\star)$.

(iii) By theorem XV.15, $\Theta < \Theta_\dagger$ implies weak coupling (47), which in turn makes the linearised OU rate $\eta\kappa_k$ positive definite and the fluctuations $\delta\lambda_k$ sub-Gaussian and bounded in L^2 by $O(1/\sqrt{D})$ (theorem XV.3). The nonlinear corrections to the OU coefficients are then $O(\delta\lambda_k^2)$, hence $O(1/D)$, negligible against the $O(1/\sqrt{D})$ fluctuations. The Lipschitz condition is therefore verified for all $\tau\in[0,T^\star)$ by induction on the resolution front: the linearisation is consistent wherever alignment holds, and alignment holds by definition on the window.

With the Lipschitz hypothesis in hand, the propagation-of-chaos argument is standard. For a typed McKean–Vlasov system with $\mathcal{W}_2$-Lipschitz, non-autonomous interaction, the synchronous coupling between the N-particle system and N i.i.d. copies of the limit obeys $\frac{d}{d\tau}\mathbb{E}\mathcal{W}_2^2(\mu_\tau^N,\mu_\tau) \le 2\ell(\tau)\,\mathbb{E}\mathcal{W}_2^2 + O(1/N)$, and Grönwall gives (59) with factor $\exp\int_0^\tau \ell$; the heterogeneous and quantitative refinements are [31], [32]. The factor is finite precisely for $\tau < \tau_c$. The non-autonomy is immaterial—only the *time-integral* of the modulus enters—which is why the evolving kernel poses no obstruction *below* τ_c. □

Theorem XV.14 already upgrades Part 1: the rigorous closure holds not merely under *instantaneous* weak coupling but on the whole *trajectory-integrated* sub-critical interval $[0,\tau_c)$. It remains to locate τ_c relative to the Adam window $T^\star$.

Lemma XV.15 (Sub-criticality is self-maintained on the alignment window; route 2)**.** *Let $\Theta(\tau)$ be the gap-weighted alignment functional of theorem XV.7. Because the off-diagonal velocity equals $\eta\mathcal{J}$ up to $O(1)$ spectral factors and the gaps satisfy $\lambda_k - \lambda_j \le \lambda_1 = O(1)$,*

$$\|\mathcal{J}(\tau)\|_{\mathrm{op}}^2 \;\le\; \sum_{j\neq k} \mathcal{J}_{jk}^2 \;\le\; \frac{C_0}{\eta^2}\,\Theta(\tau) \qquad (C_0 = O(1) \textit{ architectural}). \tag{60}$$

Hence, redefining the operative ceiling of theorem XV.8 as

$$\Theta_\dagger := \min\Big\{ \underbrace{\Theta_{\mathrm{descent}}}_{\textit{theorem XIII.7}} \,,\; \underbrace{C_0^{-1}\,\eta^3\,(\inf_k \kappa_k)^2}_{\textit{sub-criticality}} \Big\}, \tag{61}$$

the event $\{\Theta(\tau) < \Theta_\dagger\}$ implies both *alignment and weak coupling* (47). *Consequently $\tau_c \ge T^\star$, with $T^\star$ the window of* (54) *computed at the ceiling* (61)*:* the strong-coupling phase is not reached before the alignment window closes.

Proof. Equation (60) is theorem XV.7 read backwards: $\Theta = \sum_{j\neq k} \left\langle \hat\psi_j,\, \dot K \hat\psi_k \right\rangle^2 / (\lambda_k - \lambda_j)^2 \ge (\lambda_1)^{-2}\eta^2 \sum_{j\neq k} \mathcal{J}_{jk}^2 \ge (\lambda_1)^{-2}\eta^2\,\|\mathcal{J}\|_{\mathrm{op}}^2$. With $\Theta < \Theta_\dagger \le C_0^{-1}\eta^3(\inf\kappa)^2$ this gives $\|\mathcal{J}\|_{\mathrm{op}}^2 < \eta(\inf\kappa)^2$, i.e. (47); and $\Theta < \Theta_{\mathrm{descent}}$ is alignment. Since Θ is monotone increasing along the front (theorem XV.8), the first exit from $\{\Theta < \Theta_\dagger\}$ is $T^\star$ and occurs no later than the first sub-criticality breach τ_c. □

Theorem XV.16 (The chain is unconditional on the window; for all time under a decaying schedule)**.** *On $[0, T^\star]$ the DMFT closure of Part 1 holds* without assuming weak coupling*: it is enforced by theorem XV.15 and certified by theorem XV.14. If in addition the learning-rate schedule is summable enough that the alignment budget never exhausts,*

$$\int_0^\infty \eta(s)\,ds \;\le\; \big[\, b^2(2b+3-2\gamma)\,\Theta_\dagger \,\big]^{\,b_0/(2b+3-2\gamma)} \qquad \Big(\textit{e.g. } \eta = \eta_0(1+\tfrac{\tau}{\tau_0})^{-p},\ p>1,\ \eta_0 \textit{ small}\Big), \tag{62}$$

then $T^\star = \tau_c = \infty$: the strong-coupling phase is never *reached, and the closure—hence the entire chain* (63)*—is unconditional for all training time.*

Proof. The first claim is theorem XV.15 composed with theorem XV.14 ($\tau_c \ge T^\star$ makes $[0, T^\star] \subset [0, \tau_c)$). For the second, by theorem XV.8 $\Theta(\tau) \asymp b^{-2}(2b+3-2\gamma)^{-1}(\int_0^\tau \eta)^{(2b+3-2\gamma)/b_0}$ is bounded by $\Theta_\dagger$ for all τ iff (62) holds, so $T^\star = \infty$; theorem XV.15 then gives $\tau_c = \infty$. □

Corollary XV.17 (Conditional nonlinear H1 becomes a theorem on the window)**.** *On $[0, T^\star]$ (for all time, under* (62)*), theorem XI.1 holds as a* derived *consequence, so the hypothesis of theorem XI.2 is discharged and the capacity penalty $\sum_k \lambda_k^\nu$, $\nu = 1/L$, holds for nonlinear μP as a theorem—no longer conditional on an assumed eigenbasis stability. The matching value $\nu = d^*/(2s)$ of theorem IV.3 is then a theorem in the same standing on the window.*

b) Is the residue even non-empty? Two results that shrink it to nothing the paper uses.: Before leaving the glassy tail open, I settle how much of it can be closed directly. The natural attempt—prove the potential (8) convex, so the Gibbs/NESS measure is log-concave and RSB is impossible—*fails on inspection*, and instructively: the capacity term $\mu\sum_k \lambda_k^\nu$ with $\nu = 1/L < 1$ is *concave*, $\partial_{\lambda_k}^2(\mu\lambda_k^\nu) = \mu\nu(\nu-1)\lambda_k^{\nu-2} < 0$, so V is a sum of a convex reconstruction term and a concave capacity term and is *not* convex. The non-convexity is not removable; it is the rich regime. The correct tool is weaker than convexity and still sufficient.

Lemma XV.18 (No replica-symmetry breaking in the weak-coupling phase)**.** *Each coordinate potential $g_k(\lambda) = \bar a_k^2/\lambda + \mu\lambda^\nu$, $\lambda > 0$, $\nu \in (0,1)$, is* unimodal*: $g_k \to +\infty$ at both ends and g_k' vanishes at the single point $\lambda_k^* = (\bar a_k^2/\mu\nu)^{1/(\nu+1)}$ of* (10)*, a strict global minimum—no convexity required. Consequently V is separable with a unique global minimiser. In the weak-coupling phase* (47) *the Hessian $\nabla^2 V_{\mathrm{tot}} = \mathrm{diag}(\kappa_k) + (\textit{coupling})$ is positive definite ($\inf_k \kappa_k > \|\mathcal{J}\|_{\mathrm{op}}^2/(\eta\inf_k\kappa_k)$), so V_{tot} is strictly convex on the* *relevant basin and the effective single-mode measure of theorem XV.4 has a* unique pure state*: RSB does not occur. Equivalently, RSB requires the strong-coupling phase, where the Hessian first loses a positive direction.*

Proof. The unimodality of g_k is the computation after (10): $g_k'(\lambda) = -\bar{a}_k^2/\lambda^2 + \mu\nu\lambda^{\nu-1}$ is negative near 0 and positive at ∞ (since $\lambda^{\nu-1}$ decays slower than λ^{-2}), crossing zero once. Positive-definiteness of the Hessian under (47) is theorem XV.5 read as a statement about $\nabla^2 V_{\mathrm{tot}}$ rather than $\mathcal{F}$; strict convexity on the basin gives a unique Gibbs state by standard log-concavity, hence no RSB. □

Proposition XV.19 (The source exponent is RSB-robust; route (b))**.** *Suppose RSB does occur in the strong-coupling phase, so full propagation of chaos fails and the limit measure is a non-trivial mixture of pure states. The quantity R1–R4 adjudicate is the source exponent $r = t/b$, a functional of the tail* scaling exponent b *of the spectral envelope $\bar{\lambda}_k \asymp k^{-b}$, not of its constants or overlaps. Two facts make b—hence r—survive RSB:*

(i) ***Intensive self-averaging.*** *The tail exponent is an intensive spectral observable (a free-energy-density-type quantity). Such observables concentrate even in the RSB phase, by the Guerra–Toninelli self-averaging of the free energy [34]; what fails to self-average under RSB is the overlap distribution, which does not enter r.*

(ii) ***Exponent rigidity.*** *In every pure state the tail balance is the* same*: $\eta\bar{a}_k^2\lambda_k^{-2} \approx \varepsilon\lambda_k^{-\gamma}$ ((35)) for the driven case, $\approx \eta\mu\nu\lambda_k^{\nu-1}$ ((10)) for the relaxed case, because V and the injection F are common to all pure states; the only pure-state-dependent term, the memory drift Γ_k, is subdominant at the deep tail (the weak/strong split concerns the head/activating mode, not the tail). The balance is exponent-rigid, so b is identical across pure states.*

Hence r concentrates on a single value regardless of RSB: full measure-level propagation of chaos is not needed *for any claim of the paper.*

Remark XV.20 (The irreducible residue: the glassy tail, and why it is a hard wall)**.** *After theorems XV.18 and XV.19 the residue is much smaller than it first appears: RSB cannot occur in weak coupling at all, and even if it occurs in strong coupling the source exponent—the only quantity R1–R4 use—is RSB-robust.*

What is still genuinely open. *The single remaining gap is full* measure-level *propagation of chaos in the regime $\tau > \tau_c$ under a non-summable schedule $\int_0^\infty \eta = \infty$: once $\int \ell$ diverges, the synchronous-coupling factor in* (59) *blows up and theorem XV.14 is silent. This is not a slack bound but a genuine wall: in a strongly coupled phase the self-consistent measure can undergo replica-symmetry breaking—multiple pure states, non-Gaussian fluctuations, aging—under which propagation of chaos is known to* fail*, and no current technique (including [31], [32]) closes it for a self-generated, feedback-evolving disorder.*

Why this residue is inessential to the paper's claims. *Two independent closures remove it from the logical dependencies: (i) by theorem XV.15 the glassy tail can only occur* after *the eigenbasis has rotated past $\Theta_\dagger$, where the diagonal source-exponent reading of section XI-E no longer certifies an operative r, so every result of section XV is unaffected; (ii) by theorem XV.19 the source exponent $r = t/b$—the scalar that R1–R4 exclusively adjudicate—self-averages to a single value regardless of whether RSB occurs, because it is an intensive tail-scaling observable immune to the overlap non-self-averaging that is the hallmark of RSB. The glassy tail can therefore affect the prefactor of the loss curve and the non-Gaussian fluctuations of the feature kernel; it cannot affect the rate exponents α, β, r. I leave it open with this comment as a hard wall, not a gap to be patched.*

Remark XV.21 (Partial progress: a one-sided bracket on the glassy prefactor)**.** *One quantitative statement can be added without touching the wall. The residue the glassy tail can still influence is the loss* prefactor *(not the exponents, by theorem XV.19). For the self-consistent single-site free energy f_{q} (quenched) controlling that prefactor, Jensen's inequality applied to the disorder average gives the one-sided annealed bound*

$$f_{\mathrm{q}} \;\le\; f_{\mathrm{ann}} := \tfrac{1}{\beta}\log \mathbb{E}_{\mathrm{disorder}} Z,$$

and the Guerra–Toninelli interpolation [34] sharpens the same side to the replica-symmetric value, $f_{\mathrm{q}} \leq f_{\mathrm{RS}} \leq f_{\mathrm{ann}}$*, with both right-hand sides computable from the weak-coupling closure. Thus the effect of a possible strong-coupling RSB on the prefactor is confined to the bounded gap* $[\,f_{\mathrm{RS}} - f_{\mathrm{q}}\,] \leq [\,f_{\mathrm{ann}} - f_{\mathrm{q}}\,]$*. This is a bracket, not a closure: it bounds* how much *the unresolved glassy phase can move the prefactor from one side, but it does not establish measure-level propagation of chaos and leaves theorem XV.20 standing. The other side (*f_{q} *from below) matches the quenched lower bound that RSB obstructs, and it remains open.*

E. Statement and status of the closure theorem

Theorem XV.22 (DMFT closure + Adam window + tight no-free-lunch)**.** *For the μP network* (4) *of finite depth L trained by* AdamW *on an $\mathcal{H}^s(\mathcal{M})$ target with intrinsic dimension d^* and source exponent t, in the limit $n \to \infty$:*

(1) ***Closure (theorem; weak coupling).*** *In the weak-coupling phase* (47)*, Γ_k of* (18) *admits a rigorous self-consistent closure* (51)*: the coloured noise is determined, not whitened (theorems XV.2 to XV.4), and the weak/strong phase boundary is preserved (theorem XV.5). The strong-coupling closure remains open (theorem XV.6) and is not used below.*

(2) ***Adam window (theorem; weak coupling, $t > \frac{1}{2}$).*** *There is an explicit horizon $T^\star(d^*, s, L, \eta(\cdot))$,* (54)*, on which theorem XI.1 is* derived *(theorem XV.8) and, for $T^\star > \tau_{\mathrm{relax}}$, $r_{\mathrm{Adam}} > \frac{1}{2}$ holds* (55) *(theorem XV.10).*

(3) ***Tightness (theorem; unconditional).*** *At fixed entropy-production rate σ the maximal r is r_{drive}, attained by injection (A), and $\beta(r_{\mathrm{drive}})$ is the tight ceiling (theorems XV.12 and XV.13).*

Remark XV.23 (What is proved, what remains open)**.** *Part 3 is unconditional and joins R1/R4 in full standing. Parts 1 and 2 are theorems on the weak-coupling phase* (47)*—an* intrinsic, checkable *condition on bare coupling and curvature, not a hidden assumption—and the earlier hypotheses "shell self-averaging" and "drift–gap budget" are discharged into theorems XV.2, XV.3 and XV.8, the first resting only on the Gaussianity of theorem III.1. The former residue—the strong-coupling closure—is itself now closed on the alignment window by theorems XV.14 to XV.16: sub-criticality is self-maintained wherever alignment holds, so the closure is unconditional on $[0, T^\star]$, and unconditional for all time under a sufficiently decayed schedule* (62)*; theorem XV.17 then promotes the conditional nonlinear H1 (theorem XI.2) to a theorem on the window, free of theorem XI.1. The sole irreducible residue is the post-window glassy tail (theorem XV.20): a strongly coupled phase where replica-symmetry breaking can defeat propagation of chaos, which no current tool closes for self-generated disorder. It is asymptotically inert—R1 (theorem VI.1) pins the $D \to \infty$ exponent regardless—so it bears on the prefactor, never the rate. A by-product is a correction to theorem XI.1: the load-bearing functional is the gap-weighted Θ of* (52)*, not $\sum_k |\Gamma_k|$, which is why the window is finite at constant learning rate.*

F. The chain (19)*, closed where decidable*

$$\underbrace{\text{weak coupling (47)}}_{\text{checkable}} \xRightarrow{\text{theorems XV.2 and XV.3}} \underbrace{\text{rigorous closure of } \Gamma_k}_{\text{Part 1}}$$
$$\xRightarrow{\text{theorem XV.8}} \underbrace{\textit{theorem XI.1} \text{ on } [0, T^\star]}_{\text{Part 2 bridge}} \tag{63}$$
$$\xRightarrow{\text{theorem XV.10}} r_{\mathrm{Adam}} > \tfrac{1}{2}$$

capped tightly at $\beta(r_{\mathrm{drive}}(\sigma))$ by Part 3. The exponent algebra is consistent with section XI-A: $r_{\mathrm{Adam}} \downarrow \frac{1}{2}$ as $\gamma_{\mathrm{eff}} \uparrow 1 - \frac{1}{2t}$ recovers β_0; the $t > \frac{1}{2}$ threshold is the same rough/smooth divide as $L = 2s/d^*$ in theorem IV.4; and the $D \to \infty$ asymptote on the worst-case ball is still β_0, so R1 is sharpened, not breached (theorem XIII.8). The one node (63) does not reach is the strong-coupling branch, which is the outer boundary of the programme rather than a gap in it.

XVI. Synthesis: a Spectral Theory of Neural Scaling

The three preceding extensions opened exactly the doors that section X proved must be opened if anything is to exceed the stationary barrier: section XII moved the *exponent* through reduced effective dimension $d^*_{\rm loc} < d^*$; section XIII moved the *prefactor and the time-to-asymptote* through transient, driven $r > \frac{1}{2}$; and section XIV carried both onto cross-entropy and discrete data, making d^* measurable. I now assemble them. The claim of this section is not a new theorem but a *structure*: the two exits are the two independent coordinates of the loss curve, the forward maps invert into a derived scaling law, and the empirical phenomenology (additive power law, shoulders, the data/parameter allocation) reads off that structure. I keep the discipline of section XI-A: what is assembled algebra is marked theorem, and the two claims that *couple* the coordinates are marked conjecture.

A. Unifying the two exits: the scaling phase diagram

Introduce two order parameters. The *structural* parameter

$$\rho := \frac{d^*_{\rm loc}}{d^*} \in (0,1] \qquad (\rho = 1 \text{ Sobolev/unstructured},\ \rho < 1 \text{ compositional}), \tag{64}$$

and the *dynamical* parameter $r_{\max} \geq \frac{1}{2}$, the largest operative source exponent the optimiser sustains (theorem XIII.6, eq. (36), capped by theorem XIII.7). Theorem XIII.8 already named these complementary; the content here is that they are *orthogonal* controls.

Proposition XVI.1 (Orthogonal decomposition of the loss curve)**.** *On a target with effective dimension $d^*_{\rm loc}$ trained by an optimiser with sustained $r_{\max}$, the test loss obeys, for D in the scaling window,*

$$\mathbb{E}L(N,D) \asymp \underbrace{N^{-2s/d^*_{\rm loc}}}_{\text{set by } \rho} + \underbrace{c(r_{\max})\, D^{-\beta(r_{\max};d^*_{\rm loc})}}_{\text{prefactor set by } r_{\max}} + L_0, \qquad \beta(\tfrac{1}{2}; d^*_{\rm loc}) = \beta_0(d^*_{\rm loc}) = \frac{2s}{2s + d^*_{\rm loc}}, \tag{65}$$

in which ρ fixes the asymptotic exponents $\alpha = 2s/d^*_{\rm loc}$ *and $\beta_0(d^*_{\rm loc})$ (the part that survives $D \to \infty$), while $r_{\max}$ fixes the* prefactor $c(r_{\max})$ *and the pre-asymptotic exponent $\beta(r_{\max}; d^*_{\rm loc}) > \beta_0(d^*_{\rm loc})$ realised at finite D on easier-than-worst-case targets, amortising to $\beta_0(d^*_{\rm loc})$ as $D \to \infty$.*

Sketch. The exponent statement is theorem XII.6 (approximation at $d^*_{\rm loc}$) and theorem V.1 and eq. (13) read at dimension $d^*_{\rm loc}$ (estimation). The prefactor statement is theorem XIII.7 and theorem XIII.8: a driven NESS realises $\beta(r_{\max}) > \beta_0$ over its finite budget, then relaxes. The two enter (65) additively because they act on disjoint factors—$d^*_{\rm loc}$ on the N-term through the width (26), $r_{\max}$ on the D-term through the operative kernel—and neither alters the other's exponent. □

Theorem XVI.1 foliates the $(\rho, r_{\max})$ quadrant into four regimes, table I. Regime I is the entire parent paper; II is the home of the feature-learning-improves-scaling phenomenon of [11]; III is the driven/Adam prefactor gain of section XIII; IV is their superposition.

So far the axes are independent *as properties of the asymptote*. The sharper question of section XII-E is whether they are independent *as realised by SGD on a hierarchical target*. I believe they are not, and state the coupling as a conjecture, in the discipline of (H-leap) from theorem XII.7.

Conjecture XVI.2 (The staircase couples the two axes)**.** *On a compositional target* (25) *with levels of increasing local dimension $d_1 < \cdots < d_L = d^*_{\rm loc}$, first-order training cannot reach the $d^*_{\rm loc}$-asymptote without traversing a sequence of saddle escapes [20], one per constituent. Each escape is a* transient $r > \frac{1}{2}$ episode *localised to the constituent being resolved (theorem XIII.9): a structured trajectory through the diagram visits regime IV level by level, climbing ρ from d_1/d^* toward 1 as it resolves levels and spiking*

TABLE I

THE SCALING PHASE DIAGRAM. $\rho = d^*_{\rm loc}/d^*$ (STRUCTURAL AXIS); $r_{\max}$ (DYNAMICAL AXIS). ONLY $\rho < 1$ MOVES THE ASYMPTOTIC EXPONENT; $r_{\max}$ MOVES THE PREFACTOR AND THE APPROACH, AMORTISING AS $D \to \infty$.

	$r_{\max} = \frac{1}{2}$ (stationary)	$r_{\max} > \frac{1}{2}$ (driven/transient)
$\rho = 1$	**I.** Sobolev barrier: $\alpha = 2s/d^*,\ \beta = \beta_0(d^*)$	**III.** prefactor boost; asymptote still $\beta_0(d^*)$
$\rho < 1$	**II.** exponent gain: α, β governed by $d^*_{\rm loc}$	**IV.** both: exponent at $d^*_{\rm loc}$, accelerated finite-D approach

r at each escape, before relaxing onto the regime-II asymptote set by $d^*_{\rm loc} = \max_i d_i$. Hence hierarchy (the ρ-axis) induces *the transient dynamics (the $r_{\max}$-axis): the empirical staircase is the projection of this trajectory, and the two "kinds" of staircase distinguished in theorem XII.11—resource crossing (r flat) and saddle escape (r rising)—are the asymptotic and transient shadows of the* same *multi-level object.*

This is the unification the phase IV brief asked for: the *coordinates* are orthogonal (theorem XVI.1), but the *trajectory* a structured dataset forces is pinned to a staircase that visits both. Theorem XVI.2 is decidable by the multi-level reading of theorem X.1: measure $\hat{r}(t)$ across each shoulder; flat-at-$\frac{1}{2}$ segments punctuated by rising spikes confirm it, a uniformly flat $\hat{r}$ refutes it (the levels would then be resolved without saddles).

B. The inverse problem: a derived scaling law

The forward maps are now complete: structure $\to d^*_{\rm loc}$ (theorem XII.6); depth–smoothness $\to L^\star = 2s/d^*_{\rm loc}$ (theorem XII.12); loss geometry $\to (\alpha, \beta_0, L_0)$ (theorems VIII.1 and XIV.4 and eq. (38)). Inverting them answers the design question.

Theorem XVI.3 (Derived compute-optimal scaling, conditional on measurement)**.** *Let a corpus have measured invariants $(d^*, d^*_{\rm loc}, s, L_0)$ via theorem XIV.8, with $L_0 = \mathbb{H}(y \mid x)$. Under theorem XIV.5 and the non-degeneracy of section XIV-F, the compute-optimal allocation along $C = ND$ for the additive loss* (65) *is, in the stationary asymptote,*

$$\boxed{\alpha = \frac{2s}{d^*_{\rm loc}}, \quad \beta = \frac{2s}{2s + d^*_{\rm loc}}, \quad L^\star = \max\left\{\frac{2s}{d^*_{\rm loc}},\, L_{\rm hier}\right\}, \quad N^\star \propto C^{\frac{d^*_{\rm loc}}{2(s+d^*_{\rm loc})}}, \quad D^\star \propto C^{\frac{2s+d^*_{\rm loc}}{2(s+d^*_{\rm loc})}}} \tag{66}$$

with $N^\star D^\star = C$ identically, the knee at $D_{\rm cross} = N^{(2s+d^*_{\rm loc})/d^*_{\rm loc}}$, and the data-heavy tilt $\alpha - \beta = 4s^2/[d^*_{\rm loc}(2s + d^*_{\rm loc})] > 0$. Regimes I/II are served by a stationary optimiser; in regimes III/IV a driven optimiser (Adam, section XIII-E) improves the finite-C prefactor through the transient $r_{\max}$, without changing (66)*'s exponents.*

Proof. Every entry is a parent identity read at $d^*_{\rm loc}$: α from (14), β from (13) at $r = \frac{1}{2}$, $L^\star$ from (30) intersected with the representational bound of theorem XII.13. The allocation minimises $c_1 N^{-\alpha} + c_2 D^{-\beta}$ under $ND = C$: the Lagrangian gives $N^\star \propto C^{\beta/(\alpha+\beta)}$, $D^\star \propto C^{\alpha/(\alpha+\beta)}$, and substituting (α, β) yields the boxed exponents (checked symbolically; they sum to 1). Replacing $d^*_{\rm loc} \to d^*$ recovers section IX verbatim, the unstructured case. □

Theorem XVI.3 is the *proven—conditional—version of* (1): it predicts the exponents and the allocation from measurable data invariants rather than fitting them. Its limits are exactly those of the parent paper. First, only *exponents* are derived; the constants c_1, c_2 are not, so absolute loss is not predicted. Second, the internal-consistency caveat of section XI-A carries over with $d^* \to d^*_{\rm loc}$: matching $\alpha \approx 0.34$ forces $s/d^*_{\rm loc} \approx 0.17$ and $b_0 \approx 0.34 < 1$, outside the trace-class window in which the β derivation is valid—the model lands in the right ballpark ($\beta \approx 0.25$ vs. observed 0.28) but does not cleanly apply, and this gap is data, not noise. Third, (66) predicts a *data-heavy* optimum ($D^\star$ exponent $> \frac{1}{2} > N^\star$ exponent, since $\alpha > \beta$), the asymmetry also reported by [10], not the near-balanced $a \approx b \approx \frac{1}{2}$ of the empirical compute-optimal fit.

Conjecture XVI.4 (Near-balance is a transient-$r_{\max}$ signature)**.** *The discrepancy in the third limit is itself diagnostic. Theorem XVI.3's data-heavy tilt is the* stationary *asymptote; over the finite-compute window in which the empirical law is fit, a driven optimiser sustains $\beta(r_{\max}) > \beta_0$ (theorems XIII.10 and XVI.2), lifting the operative β toward α and pulling the optimum toward balance. The prediction has a sign: as C grows past the point where the transient budget of theorem XIII.7 is amortised, the compute-optimal allocation should drift* data-heavy, *toward* (66). *An observed drift confirms that the empirical near-balance is a transient, not stationary, feature—and closes the loop between the two axes.*

C. The central thesis

The four phases assemble into one statement, which I offer as the organising thesis of the programme rather than as a theorem.

> *A neural scaling law is not an empirical relation between loss and compute; it is a structural law governing how compute is allocated across the hierarchical levels of the data. The stationary Sobolev barrier $\beta_0(d^*)$ is the fundamental limit of the first (unstructured) level. Each subsequent shoulder of the empirical curve is a transition to a deeper level, governed by its own effective dimension d_i and its own sustainable transient $r_{\max,i}$; the asymptotic exponent of the whole is set by the hardest resolved level, $d^*_{\text{loc}} = \max_i d_i$, and the visible curvature is the trajectory theorem XVI.2 traces across the phase diagram table I.*

The supporting structure is the four phases, each at its proper grade. **(1) The barrier** (R1–R4, this paper, with R3 closed for the diagonal/deep-linear class and conditional for nonlinear μP) fixes the first-level limit and proves it cannot be stationarily exceeded. **(2) Hierarchical data** (section XII) supplies the ρ-axis: the only route to a sub-Sobolev *exponent*, via $d^*_{\text{loc}} < d^*$, with the matched depth $L^\star = 2s/d^*_{\text{loc}}$. **(3) Non-stationary dynamics** (section XIII) supplies the $r_{\max}$-axis: the only route to a finite-D *prefactor* gain, at a quantified thermodynamic cost, reverting asymptotically. **(4) Real data** (section XIV) makes the invariants measurable and the contact with language falsifiable, and identifies the floor L_0 with the Bayes conditional entropy. The synthesis (theorems XVI.1 and XVI.3) shows these are one object: a derived, measurement-conditional scaling law whose exponents are data invariants and whose deviations from the empirical fit are themselves predictions (theorem XVI.4).

D. Status of section XVI and the one decisive experiment

a) Theorems (assembled, under stated hypotheses).: Theorem XVI.1 (orthogonal decomposition, by theorems XII.6 and XIII.8) and theorem XVI.3 (the derived allocation, by the parent identities read at d^*_{loc}). Both reduce to section IX at $\rho = 1$, and the boxed exponents in (66) have been checked symbolically to sum to unity and to recover theorem VIII.3's gap with $d^* \to d^*_{\text{loc}}$.

b) Conjectures (named, open).: **(C-staircase)** theorem XVI.2: hierarchy forces a regime-IV trajectory, so the ρ- and $r_{\max}$-axes are dynamically coupled though asymptotically orthogonal; it rests on (H-leap) of section XII-G and the DMFT closure of section XI-C. **(C-balance)** theorem XVI.4: the empirical near-balanced allocation is a transient-$r_{\max}$ effect that should drift data-heavy at large C; it rests on (C-staircase) and the Adam hypothesis theorem XIII.10. Neither bears on R1–R4.

c) The one decisive experiment.: The entire edifice is adjudicated by a single extended measurement, the union of every test in the paper. Across a model's training, on data of independently measured $(d^*, d^*_{\text{loc}}, s)$ (theorem XIV.8) with the double-d^* consistency check (theorem XIV.10): track the operative source exponent $\hat{r}(t)$ and the binding effective dimension shoulder by shoulder (theorems X.1 and XII.11). The thesis predicts, jointly: (i) flat $\hat{r} = \frac{1}{2}$ plateaus punctuated by rising spikes at each shoulder (C-staircase); (ii) an asymptotic exponent set by the largest resolved d_i (theorem XII.6); (iii) a compute-optimal allocation drifting from balance toward the data-heavy (66) as C grows (C-balance); and (iv) a fitted floor L_0 equal to the measured conditional entropy (section XIV-F). A theory that survives all four becomes the spectral theory of neural scaling; a failure of any one localises precisely which phase is wrong. That falsifiability—each claim pinned to a measurement, none promoted above the grade its proof earns—is the discipline the programme carries throughout.

d) Decisive-experiment protocol (specification, programme grade).: For reproducibility we specify the experiment concretely rather than describe it.

(P1) **Data.** A corpus with independently measured invariants $(d^*, d^*_{\text{loc}}, s, L_0)$ via theorem XIV.8, passing the double-d^* consistency check (theorem XIV.10); record the per-level dimensions $d_1 < \cdots < d_L$.

(P2) **Models.** A μP depth/width ladder with parameter counts N spanning at least 1.5 decades, depth fixed at $L^\star = 2s/d^*_{\text{loc}}$ (theorem XII.12).

(P3) **Compute grid.** Token budgets D on a logarithmic grid straddling the predicted knee $D_{\text{cross}} = N^{(2s+d^*_{\text{loc}})/d^*_{\text{loc}}}$ (theorem XVI.3), at least three points on each side, with $C = ND$ held on iso-compute slices.

(P4) **Checkpoints.** Logarithmic cadence in training step t, ≥ 20 checkpoints per decade, dense enough to resolve each shoulder.

(P5) **Estimator.** At each checkpoint fit the feature-kernel spectral envelope $\hat{\lambda}_k(t) \asymp k^{-b(t)}$ and form the operative source exponent $\hat{r}(t) = t_{\text{eff}}(t)/b(t)$ with the instrumentation of section XI-E, tracked shoulder by shoulder together with the binding shoulder dimension.

(P6) **Predictions and falsifiers.** *(i)* $\hat{r}(t)$ shows flat $\frac{1}{2}$ plateaus punctuated by spikes $> \frac{1}{2}$ at each shoulder (C-staircase); a uniformly flat $\hat{r}$ *refutes* C-staircase. *(ii)* the asymptotic β is set by the largest resolved d_i (theorem XII.6). *(iii)* the compute-optimal allocation drifts from balance toward the data-heavy (66) as C grows past the amortisation point of theorem XIII.7 (C-balance); a stationary or model-heavy drift *refutes* C-balance. *(iv)* the fitted floor L_0 equals the measured conditional entropy (section XIV-F).

Remark XVI.5 (Why this is stated as a protocol and not reported as a result)**.** *The protocol above is at* programme *grade: the claims (i)–(iv) are predictions to be tested, not results obtained here. Two cautions apply. First, exhibiting the staircase at small, controlled scale is* not *probative: a synthetic trajectory can be* constructed *to display $\frac{1}{2}$-plateaus with rising spikes (e.g. by superposing transient bumps on a flat baseline), and such a construction confirms nothing, because it assumes the very structure it appears to show. Second, the discriminating content of the experiment lives at* LLM scale on natural data*, where $(d^*, d^*_{\text{loc}}, s)$ are not designed but measured and the allocation drift (iii) can be observed over a real compute window; that run is the frontier deliverable and is* not *performed in this work. We record the specification so that the prediction is falsifiable, while marking the large-scale test as the open decisive experiment.*

XVII. Conclusion

I have replaced an attempt to derive the Chinchilla law with a map of the regime in which stationary feature learning operates. The stationary attractor is pinned to the Sobolev minimax rate β_0 (R1), self-organises to the critical source exponent $r = \frac{1}{2}$ (R2), inherits a capacity penalty whose exponent is set by depth (R3), and gains nothing from adaptivity on a Sobolev target, giving $\alpha = 2s/d^* > \beta_0$ (R4). The empirical scaling law sits outside this regime: its feature-learning content, if any, must come from compositional structure ($d^*_{\text{loc}} < d^*$) or transient alignment. Scaling is thereby recast as a transition between data-limited and architecture-limited regimes, with the stationary barrier marking exactly what must be exceeded—and how—to account for what is observed.

The status is now consolidated (section XI): R1–R4 are theorems whose exponent algebra has been verified symbolically; the single remaining gap in the *form* of H1 is isolated into one named hypothesis, the eigenbasis-stability bound theorem XI.1; that hypothesis is identified with the non-perturbative control of Γ_k, so the two open programmes collapse into the single dependency chain (19); and the decisive test that adjudicates stationary versus transient kernels has been implemented and its inference logic validated.

What remains is to earn theorem XI.1 dynamically rather than assume it—the one step that would turn the conditional nonlinear H1 into an unconditional theorem. Section XV takes that step in the weak-coupling phase: it discharges theorem XI.1 on an explicit, spectral-flow-derived window, establishes the corresponding DMFT closure and a tight no-free-lunch frontier in the entropy-production budget, and isolates the strong-coupling closure as the sole remaining residue—one that, because the Adam elevation is an external-injection effect, does not lie on the path to the nonlinear H1.

REFERENCES

[1] J. Hoffmann *et al.*, "Training compute-optimal large language models," *arXiv:2203.15556*, 2022.

[2] J. Kaplan *et al.*, "Scaling laws for neural language models," *arXiv:2001.08361*, 2020.

[3] B. Bordelon, A. Canatar, and C. Pehlevan, "Spectrum dependent learning curves in kernel regression and wide neural networks," in *ICML*, 2020, pp. 1024–1034.

[4] A. Caponnetto and E. De Vito, "Optimal rates for the regularized least-squares algorithm," *Found. Comput. Math.*, vol. 7, no. 3, pp. 331–368, 2007.

[5] A. B. Tsybakov, *Introduction to Nonparametric Estimation*. Springer, 2009.

[6] G. Yang and E. J. Hu, "Tensor programs IV: Feature learning in infinite-width neural networks," in *ICML*, 2021, pp. 11727–11737.

[7] G. Yang, D. Yu, C. Zhu, and S. Hayou, "Tensor programs VI: Feature learning in infinite-depth neural networks," *arXiv:2310.02244*, 2023.

[8] A. Jacot, "Implicit bias of large depth networks: a notion of rank for nonlinear functions," in *ICLR*, 2023; *arXiv:2209.15055*.

[9] B. Bordelon and C. Pehlevan, "Self-consistent dynamical field theory of kernel evolution in wide neural networks," in *NeurIPS*, vol. 35, 2022.

[10] B. Bordelon, A. Atanasov, and C. Pehlevan, "A dynamical model of neural scaling laws," in *ICML*, 2024, pp. 4345–4382; *arXiv:2402.01092*.

[11] B. Bordelon, A. Atanasov, and C. Pehlevan, "How feature learning can improve neural scaling laws," *J. Stat. Mech.*, 2025; *arXiv:2409.17858*.

[12] S. Gunasekar, B. Woodworth, S. Bhojanapalli, B. Neyshabur, and N. Srebro, "Implicit regularization in matrix factorization," in *NeurIPS*, 2017.

[13] S. Arora, N. Cohen, W. Hu, and Y. Luo, "Implicit regularization in deep matrix factorization," in *NeurIPS*, 2019.

[14] B. Woodworth *et al.*, "Kernel and rich regimes in overparametrized models," in *COLT*, 2020, pp. 3635–3673.

[15] R. A. DeVore, R. Howard, and C. Micchelli, "Optimal nonlinear approximation," *Manuscripta Math.*, vol. 63, pp. 469–478, 1989.

[16] R. A. DeVore, "Nonlinear approximation," *Acta Numer.*, vol. 7, pp. 51–150, 1998.

[17] Y. Yao, L. Rosasco, and A. Caponnetto, "On early stopping in gradient descent learning," *Constr. Approx.*, vol. 26, no. 2, pp. 289–315, 2007.

[18] F. Bauer, S. Pereverzev, and L. Rosasco, "On regularization algorithms in learning theory," *J. Complexity*, vol. 23, no. 1, pp. 52–72, 2007.

[19] T. Poggio, H. Mhaskar, L. Rosasco, B. Miranda, and Q. Liao, "Why and when can deep—but not shallow—networks avoid the curse of dimensionality: a review," *Int. J. Autom. Comput.*, vol. 14, pp. 503–519, 2017.

[20] E. Abbe, E. Boix-Adsera, T. Brennan, G. Bresler, and V. Niles-Weed, "SGD learning on neural networks: leap complexity and saddle-to-saddle dynamics," in *COLT*, 2023.

[21] L. Hörmander, "The spectral function of an elliptic operator," *Acta Math.*, vol. 121, pp. 193–218, 1968.

[22] S. Mandt, M. D. Hoffman, and D. M. Blei. Stochastic gradient descent as approximate Bayesian inference. *J. Mach. Learn. Res.*, 18:1–35, 2017.

[23] P. Chaudhari and S. Soatto. Stochastic gradient descent performs variational inference, converges to limit cycles for deep networks. In *Int. Conf. on Learning Representations (ICLR)*, 2018.

[24] M. I. Freidlin and A. D. Wentzell. *Random Perturbations of Dynamical Systems*, 3rd ed. Springer, 2012.

[25] L. Onsager and S. Machlup. Fluctuations and irreversible processes. *Phys. Rev.*, 91:1505–1512, 1953.

[26] D. P. Kingma and J. Ba. Adam: A method for stochastic optimization. In *Int. Conf. on Learning Representations (ICLR)*, 2015.

[27] M. Belkin and P. Niyogi, "Convergence of Laplacian eigenmaps," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 19, 2007, pp. 129–136.

[28] N. García Trillos, M. Gerlach, M. Hein, and D. Slepčev, "Error estimates for spectral convergence of the graph Laplacian on random geometric graphs toward the Laplace–Beltrami operator," *Found. Comput. Math.*, vol. 20, pp. 827–887, 2020.

[29] E. Facco, M. d'Errico, A. Rodriguez, and A. Laio, "Estimating the intrinsic dimension of datasets by a minimal neighborhood information," *Sci. Rep.*, vol. 7, art. 12140, 2017.

[30] E. Levina and P. J. Bickel, "Maximum likelihood estimation of intrinsic dimension," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 17, 2005, pp. 777–784.

[31] D. Lacker, "Hierarchies, entropy, and quantitative propagation of chaos for mean field diffusions," *Probab. Math. Phys.*, vol. 4, no. 2, pp. 377–432, 2023.

[32] F. Delarue, D. Lacker, and K. Ramanan, "From the master equation to mean field game limit theory: a central limit theorem," *Electron. J. Probab.*, vol. 24, art. 51, 2019.

[33] A. Atanasov, B. Bordelon, and C. Pehlevan, "Neural networks as kernel learners: the silent alignment effect," in *Int. Conf. on Learning Representations (ICLR)*, 2022; *arXiv:2111.00034*.

[34] F. Guerra and F. L. Toninelli, "The thermodynamic limit in mean field spin glass models," *Comm. Math. Phys.*, vol. 230, no. 1, pp. 71–79, 2002.

[35] A. Bovier, M. Eckhoff, V. Gayrard, and M. Klein, "Metastability in reversible diffusion processes I: Sharp asymptotics for capacities and exit times," *J. Eur. Math. Soc.*, vol. 6, no. 4, pp. 399–424, 2004.